Inverse-and-Edit: Simple and Effective Framework for Fast Image Editing

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Figure 1: **Qualitative comparison on Z-Image-Turbo (flow-based)** against RF-Edit, FlowEdit, FireFlow, and UniEdit-Flow (inference time below each column; one edit per row). Our method applies each edit while preserving structure, and is the fastest editor at 3.18 s.

Abstract

Diffusion-based text-to-image models have recently been distilled into few-step samplers and fast generators, reducing inference cost while maintaining high image quality. However, text-guided editing with such accelerated backbones remains challenging: inversion in a small number of steps is unreliable, local errors accumulate along the trajectory, and existing fast-editing methods typically rely on backbone-specific inversion procedures, which limits their transferability. In this work, we propose a simple and universal framework for text-guided image editing with arbitrary accelerated diffusion

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and flow-matching models. Instead of optimizing local, per-timestep objectives, we directly minimize a global editing objective over the entire inversion-generation trajectory. The framework fine-tunes only 1.5-7 % of the base-model parameters through two lightweight LoRA adapters (forward and backward), trained on fewer than 10 K triplets from the publicly available HQ-Edit dataset. We validate our method on three architecturally distinct backbones: consistency-distilled SD1.5 LCM-Dreamshaper, adversarial-distilled SDXL-Turbo, and rectified-flow-distilled Z-Image-Turbo. On PIE-Bench, our method achieves the strongest editing alignment within each backbone family while ensuring a strong level of content preservation. Inference adds no extra steps beyond the standard inversion and editing passes.

1 Introduction

Diffusion-based text-to-image models [15, 35] have become a standard tool for high-fidelity image synthesis. Beyond generation, a central downstream capability is text-guided image editing: given an input image and a target prompt, the model should apply prompt-consistent modifications while keeping unrelated content (identity, layout, background) unchanged. Achieving this balance—strong editability with faithful preservation—typically requires carefully controlling the sampling trajectory.

Editing with diffusion pipelines achieves high image quality at the cost of significant inference time. To address this latency, recent work explores distilled diffusion models. The backbones of such methods can be broadly categorized into two groups: consistency-based [30, 36] and fast generator-based approaches [32, 46]. Consistency-based methods preserve a solver-based view of diffusion dynamics by optimizing few-step solvers for the underlying differential equation (e.g., Latent Consistency Models [26], Hyper-SD [30], Invertible Consistency Distillation [37]), whereas generator-based models (e.g., SDXL-Turbo [33]) train a neural network to map standard Gaussian noise z directly to high-quality images in a few steps.

Despite the importance of editing speed, most methods were originally formulated for full-step models, where editing proceeds in two stages: inversion recovers a terminal latent x_T that reconstructs the input x_0 under y_{src}, and editing re-denoises that latent under y_{trg}. This works because adjacent states are local ($x_t \approx x_{t-1}$), making DDIM-style inversion accurate, so full-step editors focus on the generation stage, steering it via attention or guidance (e.g., Prompt-to-Prompt [14], MasaCTRL [2], Guide-and-Rescale [40]). As distillation progressed, several works adapted full-step editing to accelerated backbones (e.g., InfEdit [45], TurboEdit [7], iCD [37], GNRi [31]). However, inversion becomes the bottleneck: few-step models traverse the noise range in large jumps, so $x_t \approx x_{t-1}$ breaks down and small inversion errors compound into poor reconstruction or drifting edits. Existing fast editors handle this in backbone-specific ways, such as replacing inversion, adding per-image latent optimization, or jointly training a dedicated inversion model (Sec. 2). These either erode the speedup or couple tightly to one distillation recipe, making transfer non-trivial.

A core limitation of prior accelerated editors is that they typically optimize "local" surrogates (e.g., enforcing step-wise consistency) and hope this transfers to the *global* objective of the full pipeline: accurate reconstruction under the source prompt and faithful editing under the target prompt. In few-step backbones, however, the trajectory consists of large jumps, so small mismatches quickly compound and the final edited image can drift even if each step is locally reasonable. Our key motivation is that accelerated models make it feasible to **differentiate through the entire inversion–generation procedure** and optimize

what we ultimately care about. We therefore treat accelerated editing as a **single end-to-end computation graph**. Given an image x_0, a source prompt y_{src}, and a target prompt y_{trg}, we (i) invert x_0 to a terminal latent x_T under y_{src}, and (ii) denoise x_T under y_{trg} to obtain the edited result $\hat{x}_0$. Instead of enforcing local consistency between neighboring states, we fine-tune two lightweight LoRA adapters—**forward** (for inversion/noising) and **backward** (for generation/denoising)—by backpropagating through the full inversion–generation pipeline. Our training objective explicitly encodes three competing requirements of fast editing: **(1) reconstruction/content preservation** via a perceptual reconstruction loss, **(2) edit strength and prompt faithfulness** via a directional CLIP objective between $(x_0, \hat{x}_0)$ and $(y_{\text{src}}, y_{\text{trg}})$, and **(3) well-posed inversion** via a prior-matching regularizer that prevents the inverted latent from becoming under-noised. Crucially, the short trajectory of few-step backbones makes this end-to-end optimization practical, and the resulting editor remains **backbone-agnostic**: the same recipe applies across distilled diffusion samplers, fast generators, and flow-based models without adding extra inference steps at test time.

In this work, we make three main contributions:

1. **Joint inversion–generation optimization for accelerated editing.** We fine-tune two lightweight LoRA adapters (forward for inversion, backward for generation) by back-propagating through the full inversion–generation cycle of a frozen few-step backbone, with a terminal-prior regularizer that closes the under-noising shortcut of this joint objective.

2. **Backbone- and distillation-agnostic recipe across U-Net and DiT.** The same recipe applies unchanged across three distinct distillation paradigms (consistency-distilled SD1.5, adversarially distilled SDXL-Turbo, and distribution-matching-distilled Z-Image-Turbo) and two architecture families (U-Net and a single-stream DiT), with no backbone-specific inversion machinery and no extra inference steps at test time.

3. **Strong quality–speed trade-off.** Across all three backbones our method achieves the strongest edit alignment within each backbone family at comparable or lower inference cost, and preserves more than prior editors at matched edit strength. On the flow backbone, a human preference study against competing flow-based editors corroborates these gains in preservation and overall preference.

2 Related work

Editing with full-step diffusion models. Most full-step methods begin with inversion, i.e., approximating a latent state of the source image at a chosen timestep using a pretrained model. This state then initializes a new, target prompt-conditioned sampling trajectory. Optimization-based approaches [27, 28, 42, 47] perform per-sample inversion optimization, which involves additional, computationally expensive iterations during the editing process. These methods improve inversion quality by optimizing prompt embeddings. Attention-based approaches [2, 14] demonstrate strong performance but may lack fine-grained controllability.

Editing with accelerated diffusion models. Distilled backbones reduce sampling cost but may degrade trajectory fidelity, making standard inversion less reliable, since deterministic inversion is not guaranteed under stochastic sampling or non-invertible updates.

Prior accelerated editors differ chiefly in whether, and how, they recover an editable starting point. Some strengthen inversion on the distilled backbone: on SDXL-Turbo, GNRi [31] and ReNoise [10] refine the inverted latent through per-image optimization. Others reduce reliance on explicit inversion: TurboEdit [7] alters the sampling dynamics via time-shifting and Delta Denoising Score [13] updates, while PostEdit [39] applies energy-based guidance during editing, and InstantEdit [11] pairs a backbone-specific rectified-flow inversion with inference-time latent injection. A third group avoids inversion altogether: InfEdit [45], on a DreamShaper-LCM backbone, adopts a virtual inversion that starts from generic Gaussian noise and combines MasaCTRL [2] with Prompt-to-Prompt [14], and ChordEdit [25] recasts one-step editing (SD-Turbo, SwiftBrushv2) as optimal transport between source- and target-prompt distributions via a *Chord Control Field* traversed in a single step. In contrast to fixed transport rules or attention edits, our method learns lightweight forward and backward adapters end-to-end on a frozen backbone, obtaining the editing direction from the jointly trained inversion and generation passes. Because it relies only on properties common to accelerated diffusion and flow samplers, namely a differentiable few-step sampling loop and standard adapter-injectable layers, rather than on any backbone-specific inversion or attention mechanism, the same recipe applies unchanged across U-Net and DiT backbones.

Fine-tuning-based fast editors. A separate line of work trains dedicated weights up front so that editing is fast at inference, rather than relying on per-image optimization or attention control alone.

iCD [37] trains a forward (inversion) and a reverse (generation) consistency model under a multi-boundary objective, obtaining its inversion ability through consistency distillation on the diffusion probability-flow ODE and, for editing, relying on Prompt-to-Prompt attention control. It is demonstrated only on consistency-distilled diffusion backbones (SD1.5, SDXL), leaving open whether its inversion extends to adversarially distilled or flow-matching models. Our method needs neither: we attach lightweight LoRA adapters to a frozen backbone and treat it as a black box, so the same recipe extends to the adversarial and flow-matching models iCD leaves untested.

SwiftEdit [29] takes a heavier route: on top of the one-step SwiftBrushv2 generator [6] (an SD2.1-distilled U-Net, $\sim$866M parameters), it fully trains a second U-Net of the same scale as a dedicated one-step inversion network in two stages, with an IP-Adapter-style conditioning branch and an attention-rescaling editing mechanism, yielding instant ($\sim$0.2 s) edits. The framework can be re-instantiated on other one-step generators (e.g., DMD2, SwiftBrush), but all remain U-Net-based (SD1.5 or SD2.1) and each re-pays the full two-stage training; no SDXL or flow/DiT backbone is demonstrated. Wu et al. [43] similarly train a separate reconstruction-only inversion encoder on SDXL-Turbo, editing at inference via caption manipulation. Ours is far lighter: instead of fully training a second generator-scale network in two stages, we adapt two lightweight LoRA adapters—a forward (inversion) and a backward (generation) adapter—on a single frozen backbone, jointly in one stage for just 300–600 iterations on fewer than 10K triplets (1.5–7% of the base-model parameters), with no IP-Adapter or attention control at inference (A component-by-component comparison to SwiftEdit [29] is given in Appendix A.1). The result is a single recipe that transfers unchanged across U-Net and transformer-based (DiT) backbones, including the flow-matching models neither iCD nor SwiftEdit demonstrates. Our scheme relates to reward fine-tuning (ReFL [44], DRaFT [8], ShortFT [12]), which backpropagates through the generation pass alone; we optimize the joint inversion–generation cycle, whose under-noising shortcut has

no counterpart there. Concurrent BiFM [4] learns both directions but trains from scratch, whereas we adapt a frozen pretrained backbone.

Editing with flow-based models. Recent text-to-image systems increasingly adopt flow-matching / rectified-flow objectives [23, 24], parameterizing a learned velocity field as an ODE (e.g., SD3 [8] and FLUX). Editing again hinges on inversion, which is difficult even in the many-step regime, since the learned velocity field makes naive ODE-reversal imprecise. Recent flow editors improve this in different ways: RF-Edit [41] uses a second-order solver and shares self-attention features across the inversion and editing passes; FireFlow [6] attains second-order accuracy at first-order cost in eight steps; UniEdit-Flow [17] pairs a predictor–corrector inversion with a region-aware sampler; and FlowEdit [20] avoids inversion altogether but still needs many steps (28–50 on FLUX/SD3). All operate with tens of steps rather than the few-step regime of aggressively distilled backbones such as Z-Image-Turbo, and several add inference-time machinery (feature-sharing in RF-Edit, region-aware sampling in UniEdit-Flow). Operating directly on a distilled rectified-flow backbone (Z-Image-Turbo), our method achieves the fastest inference among flow-based editing methods, requiring only a few sampling steps rather than the tens used by prior flow editors.

Instruct-based editing. A growing line of work replaces source/target prompt pairs with editing instructions, learning a direct image+instruction→edit mapping rather than inverting and regenerating (e.g., InstructPix2Pix [1], FLUX Kontext [21]). These typically require heavy fine-tuning of a large text-to-image model and large supervised datasets of edited ground-truth images. Our method needs neither: we keep a frozen backbone and train lightweight adapters on fewer than 10K prompt-pair triplets, with no edited ground-truth targets.

3 Preliminaries

Diffusion and flow models. We build on a latent text-to-image model that operates in the VAE latent space. Given an image x_0, we encode it into $z_0 = \mathrm{enc}(x_0)$. In the diffusion case, the forward process produces noisy latents $\{z_t\}_{t=1}^{T}$ via a variance schedule $\{\alpha_t\}$ with $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$: $z_t = \sqrt{\bar{\alpha}_t}\, z_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon$, $\varepsilon \sim \mathcal{N}(0, I)$. A denoiser $\varepsilon_\theta(z_t, t, y)$ (conditioned on text y through cross-attention) is trained by noise prediction,

$$\mathcal{L}(\theta) = \mathbb{E}_{t, z_0, \varepsilon}\left[\|\varepsilon_\theta(z_t, t, y) - \varepsilon\|_2^2 \right]. \tag{1}$$

At inference, generation starts from $z_T \sim \mathcal{N}(0, I)$ and applies a deterministic DDIM-style update (equivalently, an ODE discretization) to map $z_t \mapsto z_{t-1}$:

$$z_{t-1} = \sqrt{\frac{\bar{\alpha}_{t-1}}{\bar{\alpha}_t}}\, z_t + \left(\sqrt{1-\bar{\alpha}_{t-1}} - \sqrt{\frac{\bar{\alpha}_{t-1}}{\bar{\alpha}_t}}\sqrt{1-\bar{\alpha}_t} \right) \varepsilon_\theta(z_t, t, y). \tag{2}$$

Rectified-flow models replace the noise-prediction view with a learned velocity field. A network $v_\theta(z_t, t, y)$ is trained to match the velocity of a straight path between data and noise, $z_t = (1-t)\, z_0 + t\, \varepsilon$, by regressing the target $\varepsilon - z_0$; sampling then integrates the probability-flow ODE $\mathrm{d}z_t = v_\theta(z_t, t, y)\, \mathrm{d}t$ from z_T to z_0. A single deterministic Euler step over the interval $t \to t'$ gives

$$z_{t'} = z_t + (t' - t)\, v_\theta(z_t, t, y). \tag{3}$$

Although their training objectives differ, both (2) and (3) define a single deterministic step between adjacent noise levels along the sampling trajectory. Our method operates only on this step, which is what allows the same forward and backward adapters to apply to diffusion and flow backbones alike.

Distilled diffusion and flow models. Distillation cuts sampling from tens of steps to a few, $0 = t_0 < t_1 < \cdots < t_M = T$ with M small. Our experiments span three paradigms, each replacing the noise-prediction loss (1) with a few-step objective:

$$\text{CD:}\quad \mathcal{L}_{\text{CD}} = \left\| f_\theta(z_{t_m}, t_m, y) - f_{\theta^-}(z_{t_{m-1}}, t_{m-1}, y) \right\|, \tag{4}$$

$$\text{ADD:}\quad \mathcal{L}_{\text{ADD}} = \mathcal{L}_{\text{adv}}\big(D_W(\hat{x}_0)\big) + \lambda\, \mathcal{L}_{\text{dist}}(\hat{x}_0, x_0^{\text{tch}}), \tag{5}$$

$$\text{DMD:}\quad \nabla_\theta \mathcal{L}_{\text{DMD}} \propto \mathbb{E}\big[\, s_{\text{fake}} - s_{\text{real}} \,\big], \tag{6}$$

where f_θ is a consistency function (with EMA target copy f_{θ^-}) mapping a PF-ODE point to its endpoint; $\hat{x}_0$ is the student's few-step output, D_W a discriminator, $\mathcal{L}_{\text{dist}}$ a distillation term toward the teacher output x_0^{tch} with weight λ; and $s_{\text{real}} = \nabla \log p_{\text{real}}$, $s_{\text{fake}} = \nabla \log p_{\text{fake}}$ are the teacher and student scores, so the gradient drives $p_{\text{fake}} \to p_{\text{real}}$. These instantiate, respectively, consistency distillation (LCM [26], the DreamShaper-LCM we use on SD1.5), adversarial diffusion distillation (SDXL-Turbo [33], our SDXL backbone), and distribution-matching distillation ([46]; Z-Image-Turbo, our flow backbone). Whatever the objective, collapsing many steps into a few makes each transition span a large interval $\Delta_m = t_m - t_{m-1}$, so adjacent latents cease to be local,

$$z_{t_{m-1}} \not\approx z_{t_m}, \qquad \varepsilon_\theta(z_{t_m}, t_m, y) \not\approx \varepsilon_\theta(z_{t_{m-1}}, t_m, y), \tag{7}$$

and the near-locality that inversion-based editing relies on no longer holds.

Editing as inversion and generation. Both deterministic steps (2) and (3) map a latent at one noise level to the adjacent one. Iterating the inversion direction ($t_{m-1} \to t_m$) from $z_{t_0} = \text{enc}(x_0)$ defines the terminal latent, and iterating the generation direction ($t_m \to t_{m-1}$) from it reconstructs the source:

$$z_{t_m} = \text{step}(z_{t_{m-1}}; t_{m-1} \to t_m, y),\ \ m{=}1{:}M, \quad \hat{z}_{t_{m-1}} = \text{step}(\hat{z}_{t_m}; t_m \to t_{m-1}, y),\ \ m{=}M{:}1, \tag{8}$$

where $\text{step}(\cdot)$ is the backbone update (2)/(3), $z_T \triangleq z_{t_M}$, and $\hat{z}_{t_M} \triangleq z_T$. We adopt the standard two-stage pipeline: **inversion** runs the left recurrence under the **source prompt** y_{src} to obtain z_T, and **editing generation** runs the right recurrence from the same z_T under either prompt,

$$z_T = \text{Inv}(x_0, y_{\text{src}}), \quad \text{dec}\big(\text{Gen}(z_T, y_{\text{src}})\big) \approx x_0, \quad \hat{x}_{\text{trg}} = \text{dec}\big(\text{Gen}(z_T, y_{\text{trg}})\big). \tag{9}$$

Standard DDIM inversion solves (9) accurately for full-step samplers but degrades in the few-step regime (7); we therefore learn Inv and Gen jointly (as adapters f_ϕ, g_θ in Sec. 4) rather than approximate them, so z_T is the latent the generation stage is calibrated to denoise.

4 Method

For accelerated diffusion backbones, inversion is often the dominant failure mode: few-step transitions amplify small inversion errors, which then propagate into the edited result.

Prior fast editing methods frequently address this by introducing backbone-specific inversion tricks or per-image optimization, making transfer across distilled models non-trivial. Our key observation is that the small number of steps in accelerated samplers makes **end-to-end training over the entire inversion–generation loop feasible**. We therefore directly fine-tune a lightweight adaptation for editing so that (i) inversion produces a terminal latent that behaves like valid model noise, and (ii) denoising from that latent reliably reconstructs the source under y_{src} and performs the intended change under y_{trg}. This framing makes the reconstruction–edit trade-off explicit: we simultaneously enforce high-fidelity content preservation and strong, prompt-consistent edits.

We adapt a pre-trained distilled backbone D_W for editing using two lightweight LoRA parameter sets. Concretely, we instantiate **two copies of the same frozen network** with different LoRA weights: a *forward* model $F_\phi(\cdot) = D_{W+\Delta W_\phi}(\cdot)$, used only for inversion under y_{src}, and a *backward* model $G_\theta(\cdot) = D_{W+\Delta W_\theta}(\cdot)$, used for denoising under y_{src} or y_{trg}. The role of F_ϕ is to map x_0 to a terminal latent z_T that (a) preserves enough information to reconstruct the source, but (b) remains statistically compatible with the backbone's noise prior, so that few-step denoising remains prompt-sensitive. The role of G_θ is to make the denoising dynamics robust to the (learned) inversion distribution produced by F_ϕ, improving stability and edit fidelity. Training both adapters jointly aligns inversion and generation without modifying the backbone weights.

4.1 Latent representation alignment

Optimizing only the inversion mapping F_ϕ is possible but tends to be unstable: the model can exploit shortcut solutions that reconstruct well yet yield a terminal latent distribution that the (fixed) denoiser is not calibrated for, degrading editing. We therefore optimize F_ϕ and G_θ **jointly**, allowing the backward model to adapt to the learned inversion distribution while still keeping the shared backbone weights W frozen. This "two-sided" alignment makes training more sample-efficient and improves robustness across prompts and images, because inversion and generation are calibrated to each other under the same few-step schedule.

4.2 General framework for image editing

We operate in the VAE latent space: for an input image x_0 we compute $z_0 = \text{enc}(x_0)$. Our editing pipeline consists of a forward (inversion/noising) pass that maps z_0 to a terminal latent z_T, followed by a backward (generation/denoising) pass that maps z_T back to an image under a chosen prompt (Fig. 2). Since we target accelerated backbones with a small number of inference steps, we explicitly discretize the trajectory into M noise levels $0 = t_0 < t_1 < \cdots < t_M = T$ and train through the entire forward–backward loop end-to-end.

Reconstruction. We formulate reconstruction as a closed-loop process executed with a small number of steps. We discretize the diffusion trajectory into M noise levels $0 = t_0 < t_1 < \cdots < t_M = T$ consistent with the accelerated backbone. Given an input image x_0 and its source prompt y_{src}, we first encode x_0 into $z_{t_0} = z_0 = \text{enc}(x_0)$ and apply the forward (inversion/noising) model F_ϕ step-by-step:

$$z_{t_m} = F_\phi\big(z_{t_{m-1}}, t_{m-1}, y_{\text{src}}\big), \qquad m = 1, \ldots, M. \tag{10}$$

The terminal latent is defined as

$$z_T \triangleq z_{t_M} \equiv f_\phi(x_0, y_{\text{src}}), \tag{11}$$

which serves as the starting point for the backward (generation/denoising) process. To reconstruct the source, we denoise from z_T under the same prompt y_{src}. We initialize the backward chain by explicitly setting

$$\hat{z}_{t_M} \triangleq z_T, \tag{12}$$

and apply M backward steps:

$$\hat{z}_{t_{m-1}} = G_\theta\big(\hat{z}_{t_m}, t_m, y_{\text{src}}\big), \qquad m = M, \dots, 1. \tag{13}$$

We define the induced image-generation mapping of the backward model as

$$g_\theta(z_T, y) \triangleq \text{dec}(\hat{z}_{t_0}), \quad \text{where } \hat{z}_{t_0} \text{ is obtained by running (12)–(13) under prompt } y. \tag{14}$$

Thus, the reconstructed image is

$$\hat{x}_{\text{src}} = g_\theta\big(f_\phi(x_0, y_{\text{src}}), y_{\text{src}}\big), \tag{15}$$

and we optimize both adapters end-to-end with the reconstruction loss

$$\mathcal{L}_{\text{rec}}(\phi, \theta) = \text{LPIPS}(\hat{x}_{\text{src}}, x_0). \tag{16}$$

This term enforces faithful inversion and content preservation under the source prompt.

Editing. Minimizing $\mathcal{L}_{\text{rec}}$ alone admits a degenerate shortcut: the forward model can keep z_T insufficiently noised, so the backward pass reconstructs x_0 well but becomes weakly sensitive to y_{trg}, yielding near-identity outputs. We therefore add complementary terms that (i) keep the terminal latent compatible with the backbone's noise prior, preventing under-noising, and (ii) explicitly encourage prompt-consistent semantic change.

Prior matching for z_T. We regularize the terminal latent z_T to remain compatible with the backbone's noise prior $\mathcal{N}(0, I)$, using two complementary terms that constrain different statistics of $\tilde{z} = z_T / \sigma_T \in \mathbb{R}^{C \times H \times W}$.

Channel-space Gaussianity. Treating the spatial tokens of $\tilde{z}$ as samples in $\mathbb{R}^C$, we estimate the empirical mean $\mu(\tilde{z}) \in \mathbb{R}^C$ and covariance $\Sigma(\tilde{z}) \in \mathbb{R}^{C \times C}$, and penalize their KL divergence to the standard normal:

$$\mathcal{L}_{\text{KL}}(z_T) = \text{KL}\big(\mathcal{N}(\mu, \Sigma) \,\|\, \mathcal{N}(0, I)\big) = \tfrac{1}{2}\big(\text{tr}(\Sigma) + \mu^\top \mu - C - \log\det\Sigma\big). \tag{17}$$

This matches the per-channel means and the full channel covariance to the identity, driving the channels toward unit variance and mutual decorrelation.

Spatial whiteness. The channel-space statistics above are invariant to spatial permutations of $\tilde{z}$, and therefore cannot detect structured low-frequency content (smooth, content-bearing residuals) in the terminal latent. We close this blind spot with a low-frequency excess penalty. Let $\bar{z} = \text{AvgPool}_{k \times k}(\tilde{z}) \in \mathbb{R}^{C \times H/k \times W/k}$ denote the $k \times k$ average-pooled response, with $k = 8$. If $\tilde{z}$ were i.i.d. $\mathcal{N}(0, 1)$, each pooled entry would be the sample mean of

k^2 unit-variance values, hence zero-mean with variance $1/k^2$. We penalize energy in excess of this white-noise floor with a one-sided hinge:

$$\mathcal{L}_{\mathrm{LF}}(z_T) = \left[\mathbb{E}\left[\bar{z}^2\right] - 1/k^2\right]_+, \qquad [u]_+ = \max(u,0). \tag{18}$$

The two terms are complementary: $\mathcal{L}_{\mathrm{KL}}$ constrains the channel-space statistics (per-channel mean and full inter-channel covariance), while $\mathcal{L}_{\mathrm{LF}}$ constrains spatial whiteness at scale k. Both vanish in expectation when $\tilde{z} \sim \mathcal{N}(0,I)$, so neither produces gradient from typical i.i.d. fluctuations; $\mathcal{L}_{\mathrm{LF}}$ is moreover one-sided, penalizing only excess low-frequency structure and not a deficit.

Directional CLIP for target edits. To enforce that the produced change follows the target prompt, we generate an edited image by denoising from the same inverted latent but conditioning on y_{trg}:

$$\hat{x}_{\mathrm{trg}} = g_\theta\big(f_\phi(x_0, y_{\mathrm{src}}), y_{\mathrm{trg}}\big). \tag{19}$$

Let $E_I(\cdot)$ and $E_T(\cdot)$ denote CLIP image and text encoders. Following [9, 19], we define the induced image direction $\Delta E_I = E_I(\hat{x}_{\mathrm{trg}}) - E_I(x_0)$ and the desired text direction $\Delta E_T = E_T(y_{\mathrm{trg}}) - E_T(y_{\mathrm{src}})$, and maximize their alignment:

$$\mathcal{L}_{\mathrm{dCLIP}} = -\big\langle \Delta E_I, \Delta E_T \big\rangle. \tag{20}$$

Together, $\mathcal{L}_{\mathrm{rec}}$, $\mathcal{L}_{\mathrm{KL}}$, $\mathcal{L}_{\mathrm{LF}}$, and $\mathcal{L}_{\mathrm{dCLIP}}$ enforce the key trade-off in editing: preserve source-specific content while producing strong, prompt-consistent modifications.

Final Objective. We jointly optimize the forward and backward adapters on triplets $(x_0, y_{\mathrm{src}}, y_{\mathrm{trg}})$ by combining reconstruction, edit-direction alignment, and terminal-prior regularization. Using the definitions in (15) and (19), the overall objective is

$$\mathcal{L}(\phi,\theta; x_0, y_{\mathrm{src}}, y_{\mathrm{trg}}) = \mathcal{L}_{\mathrm{rec}}(\phi,\theta) + \mathcal{L}_{\mathrm{dCLIP}} + \mathcal{L}_{\mathrm{LF}}(z_T) + \lambda_{\mathrm{KL}}\,\mathcal{L}_{\mathrm{KL}}(z_T) \tag{21}$$

where $\mathcal{L}_{\mathrm{rec}}$ is given in (16), $\mathcal{L}_{\mathrm{KL}}$ in (17), $\mathcal{L}_{\mathrm{LF}}$ in (18), $\mathcal{L}_{\mathrm{dCLIP}}$ in (20), and $z_T = f_\phi(x_0, y_{\mathrm{src}})$ in (11). Each term is necessary (Table 1; full analysis in Appendix C): reconstruction alone collapses to identity, the prior terms restore prompt sensitivity, and the directional signal is essential; training only F_ϕ or G_θ is likewise unstable.

4.3 Editing inference

At test time, editing requires only the learned forward and backward adapters and does not rely on additional attention-control modules or per-image optimization. Given an input image x_0 with a source prompt y_{src} and a target prompt y_{trg}, we first compute the terminal latent

$$z_T = f_\phi(x_0, y_{\mathrm{src}}), \tag{22}$$

using the M-step forward process in Eq. (10). We then denoise from the same z_T under the target prompt and decode with the VAE. To amplify the edit, we replace the unconditional branch of classifier-free guidance with the y_{src}-conditioned prediction, following Delta Denoising Score [13], giving the guided backward step

$$\hat{z}_{t_{m-1}} = G_\theta\big(\hat{z}_{t_m}, t_m, y_{\mathrm{src}}\big) + \omega\left(G_\theta\big(\hat{z}_{t_m}, t_m, y_{\mathrm{trg}}\big) - G_\theta\big(\hat{z}_{t_m}, t_m, y_{\mathrm{src}}\big)\right), \qquad m = M, \ldots, 1, \tag{23}$$

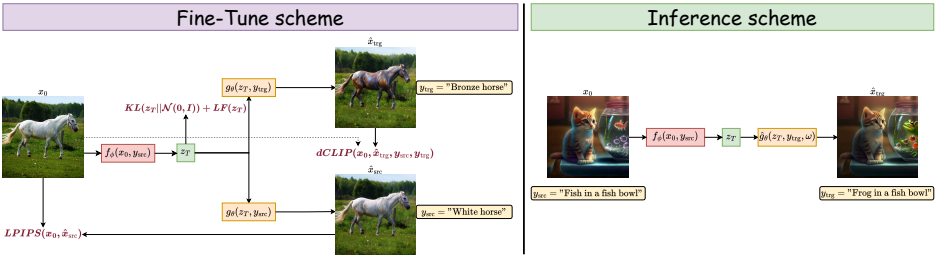


Figure 2: **Overview of our framework.** *Left (fine-tuning):* the forward adapter f_ϕ inverts x_0 to a terminal latent z_T; the backward adapter g_θ denoises z_T under y_{src} (reconstruction, scored by LPIPS) and under y_{trg} (edit, scored by directional CLIP). Both adapters are trained jointly end-to-end with a prior regularizer KL + LF on z_T and the backbone frozen (Eq. 21). *Right (inference):* the same f_ϕ inverts, then guided $\hat{g}_\theta(z_T, y_{\text{trg}}, \omega)$ produces the edit with no auxiliary machinery (Eq. 24).

Table 1: Ablation of objective terms on Z-Image-Turbo (step 600); bold marks the best *editing run*. Without $\mathcal{L}_{\text{dCLIP}}$ the model merely reconstructs the source (dCLIP 0.046, IR < 0); the full objective is best on editability and quality. LPIPS (VGG), SSIM, and PSNR are whole-image.

Method	$\text{CLIP}_{\text{wh}}\uparrow$	dCLIP↑	LPIPS↓	SSIM↑	PSNR↑	IR↑
w/o $\mathcal{L}_{\text{KL}}$	26.06	0.1350	**0.1901**	**0.8306**	21.81	0.6029
w/o $\mathcal{L}_{\text{LF}}$	26.13	0.1423	0.2071	0.8171	21.49	0.6015
Ours (full)	**26.44**	**0.1541**	0.1959	0.8296	**22.01**	**0.7383**
w/o $\mathcal{L}_{dCLIP}$ (no edit signal)	23.87	0.0457	0.0761	0.9075	25.72	–0.1320

controlled by a guidance scale ω. Like our directional CLIP objective (20), this update is driven by the source-to-target difference rather than the absolute target, keeping training and inference consistent in optimizing *relative change*. We denote the induced guided generation mapping by $\hat{g}_\theta(z_T, y, \omega)$; the original g_θ in Eq. (14) is recovered as the special case $\omega = 1$. The final edited image is

$$\hat{x}_{\text{trg}} = \hat{g}_\theta\big(f_\phi(x_0, y_{\text{src}}), y_{\text{trg}}, \omega\big). \tag{24}$$

This inference pipeline is fully deterministic and performs exactly $2M$ model calls (forward + backward), making it efficient for accelerated backbones; output diversity, when desired, is controlled via ω (Appendix D.2). Crucially, because F_ϕ and G_θ were jointly trained through the full loop with the objective in Eq. (21), the produced z_T is both prior-compatible and informative enough to preserve content, enabling strong edits without extra components.

5 Experiments

Backbones and evaluation protocol. We evaluate our framework across three families of accelerated text-to-image backbones and compare methods within the corresponding family. For the **SD family**, we include both full-step SD editors (e.g., NTI, NPI, G&R) and distilled SD variants used by accelerated editors (e.g., DreamShaper-LCM for InfEdit and our method, and the official iCD distilled model). For the **SDXL family**, all compared ac-

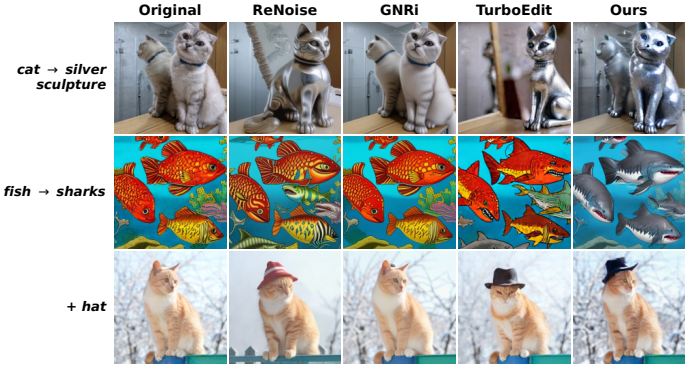


Figure 3: **Qualitative comparison on SDXL-Turbo** against ReNoise, GNRi, and TurboEdit (one edit per row). The baselines either lose input identity and detail or fail to apply the edit; our method applies each edit while preserving composition, identity, and background.

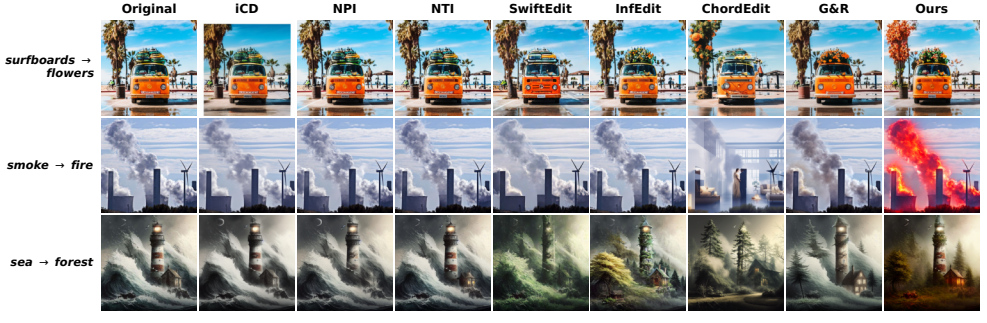


Figure 4: **Qualitative comparison on LCM-DreamShaper** against iCD, NPI, NTI, SwiftEdit, InfEdit, ChordEdit, and G&R (one edit per row). Most baselines barely edit the input or distort the scene when they do; our method produces a consistent edit per row while preserving layout, identity, and background.

celerated editors (TurboEdit, ReNoise, GNRi, and ours) are run on **SDXL-Turbo**. For the **flow / rectified-flow family**, we use **Z-Image Turbo** as the distilled flow backbone and compare against RF-based baselines RF-Edit, FlowEdit, FireFlow, and UniEdit-Flow under the same setting. This split matches our qualitative figures: Fig. 4 (DreamShaper-LCM), Fig. 3 (SDXL-Turbo), and Fig. 1 (Z-Image Turbo).

Fine-tune setup. For each backbone family (DreamShaper-LCM / SDXL-Turbo / Z-Image Turbo), we fine-tune two LoRA adapters (forward F_ϕ and backward G_θ) by minimizing the full objective in Eq. 21. The editing term uses the directional CLIP loss (Eq. 20), aligning the image embedding shift $E_I(\hat{x}_{\text{trg}}) - E_I(x_0)$ with the text shift $E_T(y_{\text{trg}}) - E_T(y_{\text{src}})$, while the reconstruction term uses LPIPS (Eq. 16, VGG-16). CLIP embeddings are computed with CLIP ViT-L/14 on 224×224 resized images. We fine-tune (ϕ, θ) on curated triplets from HQ-Edit [16], using 300 iterations (batch size 16) for SD1.5, 600 iterations (batch size 16) for SDXL-Turbo, and 600 iterations (batch size 4) for Z-Image-Turbo. Additional optimizer and hyperparameter details are provided in Appendix A.

Table 2: Baselines vs Ours under the official PIE-Bench protocol ($CLIP_{wh}$, background-masked $CLIP_{ed}$/LPIPS/SSIM/PSNR and Structure Distance $\times 10^3$); dCLIP is the directional CLIP score; ImageReward and time. Within each block, best is **bold** and second-best is underlined. Avg Rank is the mean dense-rank over the nine metrics, lower is better, with ties resolved on displayed precision.

	Editability			Preservation				Quality	Efficiency	Summary
Method	$CLIP_{wh}\uparrow$	$CLIP_{ed}\uparrow$	dCLIP↑	Struct↓	LPIPS↓	SSIM↑	PSNR↑	IR↑	Time↓ (s)	Avg Rank↓
SD-based models										
G&R	**26.47**	**22.79**	0.106	13.7	0.070	0.834	25.0	0.427	19.7	**3.33**
InfEdit	25.57	22.24	0.096	18.5	0.053	0.850	**28.0**	0.278	3.5	**3.33**
ChordEdit	25.90	22.54	0.119	29.7	0.122	0.764	22.4	0.516	3.5	5.11
SwiftEdit	24.61	21.34	0.081	16.2	0.077	0.820	24.0	–0.104	**0.2**	5.78
iCD	24.93	21.71	0.047	21.4	0.084	0.820	24.8	–0.061	1.1	5.67
NPI	24.81	21.57	0.085	13.8	0.062	0.843	26.7	0.007	9.1	4.56
NTI	24.35	21.11	0.050	**9.6**	**0.051**	**0.853**	27.9	–0.093	115.0	4.78
Ours	26.13	22.47	**0.155**	22.8	0.071	0.843	23.6	**0.573**	0.6	3.44
SDXL-based models										
TurboEdit	26.52	23.05	0.145	30.4	0.102	0.801	22.5	0.670	1.0	2.22
GNRi	24.12	20.74	0.038	**8.7**	**0.064**	**0.837**	**27.7**	–0.129	**0.6**	2.33
ReNoise	25.36	22.21	0.124	61.0	0.185	0.714	20.3	0.324	3.1	3.56
Ours	**27.17**	**23.31**	**0.178**	43.0	0.105	0.808	20.8	**0.858**	**0.6**	**1.78**
Flow-based models										
RF-Edit	25.55	22.44	0.103	21.3	0.104	0.855	25.0	0.449	30.0	3.89
FlowEdit	26.17	**22.90**	0.108	28.4	0.109	0.841	22.0	0.726	8.0	3.22
FireFlow	26.06	22.76	0.117	26.4	0.119	0.830	23.2	0.626	21.0	3.78
UniEdit-Flow	25.64	22.22	0.123	**9.3**	0.055	**0.917**	**29.7**	0.360	10.0	2.67
Ours	**26.43**	22.77	**0.154**	12.8	**0.051**	0.904	26.2	**0.738**	**3.2**	**1.44**

Inversion and editing setup. We evaluate inversion and editing on separate datasets. For the inversion experiments, we use more than 2700 high-resolution images from MS-COCO [22] for quantitative evaluation, as it provides a large amount of labeled data at varying scales, enabling reliable detection of differences between methods. We report reconstruction only for methods to which inversion is applicable, since inversion-free editors do not recover a source latent and are therefore not comparable on this task; for these reconstruction comparisons we disable classifier-free guidance for all methods. For the editing experiments, we evaluate on the full PIE-Bench benchmark of 700 images spanning its ten edit categories, with all metrics computed by the official PIE-Bench implementation [18]. Our method uses neither blend words nor any additional optimization at inference, and all baselines are run with their default settings as provided by the authors or official implementations.

5.1 Image Inversion

We assess inversion through image reconstruction, comparing within each backbone family against methods for which inversion is applicable: DDIM inversion [34] and iCD [37] for SD, SwiftEdit [29] as a one-step inversion baseline, GNRi [31], TurboEdit [7], and ReNoise [10] for SDXL-Turbo, and the flow-based methods UniEdit-Flow [17], FireFlow [6],

Table 3: **Human preference study on Z-Image-Turbo (flow-based).** Pairwise comparisons of our method against each flow editor, 50 per baseline (200 total). Each cell reports the preference score win $+\frac{1}{2}$tie (%); 50% is parity, higher favors ours. *Overall* is overall preference, *Match* is agreement with the edit prompt, and *Preserve* is fidelity to the source image.

Ours vs.	Overall ↑	Match ↑	Preserve ↑
RF-Edit	78.0	59.0	81.0
FlowEdit	76.0	58.0	81.0
FireFlow	81.0	58.0	83.0
UniEdit-Flow	70.0	59.0	70.0
Aggregate	**76.3**	**58.5**	**78.8**

Table 4: Evaluation metrics for image reconstruction across SD-, SDXL-, and flow-based methods. Best (**bold**) and second best (underlined) are computed within each block for each metric. Only methods that perform an explicit inversion are included.

SD-based methods				*SDXL-based methods*				*Flow-based methods*			
Model	MSE ↓	ImageReward ↑	LPIPS ↓	Model	MSE ↓	ImageReward ↑	LPIPS ↓	Model	MSE ↓	ImageReward ↑	LPIPS ↓
DDIM-50	0.0079	-0.023	0.279	GNRi	0.0065	0.416	0.283	UniEdit-Flow	0.0127	-0.050	0.274
iCD	0.0447	-0.497	0.611	TurboEdit	0.0470	0.427	0.483	FireFlow	0.0092	-0.102	0.327
SwiftEdit	0.0496	**0.437**	0.507	ReNoise	0.0615	0.367	0.541	RF-Edit	0.0119	-0.342	0.376
Ours	**0.0056**	0.425	**0.172**	**Ours**	**0.0061**	**0.504**	**0.186**	**Ours**	**0.0052**	**0.535**	**0.166**

and RF-Edit [41]. We measure reconstruction quality with mean-squared error (MSE), ImageReward, and LPIPS.

As reported in Table 4 (qualitative reconstructions and per-backbone figures in Appendix D.1), our method attains the lowest MSE and LPIPS in every backbone family, indicating the most faithful pixel- and perceptual-level reconstruction; on the SD backbone it surpasses even full-step DDIM inversion despite operating in the accelerated few-step regime. It also achieves the best ImageReward on the SDXL and flow backbones. The only exception is ImageReward on SD, where SwiftEdit is marginally higher (0.437 vs. 0.425) at far lower fidelity (LPIPS 0.507 vs. 0.172, MSE 0.0496 vs. 0.0056). Overall, our adapters recover the source latent more accurately than prior accelerated inversion methods while matching or exceeding their perceptual quality.

5.2 Text-guided image editing

We compare within each backbone family, matching Table 2: for SD, full-step editors (NTI [28], NPI [27], G&R [40]) and accelerated ones (iCD [37], InfEdit [45], ChordEdit [25], SwiftEdit [29]); for SDXL-Turbo, TurboEdit [7], ReNoise [10], and GNRi [31]; and for the flow family, RF-Edit [41], FireFlow [6], UniEdit-Flow [17], and FlowEdit [20].

Qualitative evaluation. We show representative results for each backbone family: LCM-DreamShaper [26] in Figure 4, SDXL-Turbo [33] in Figure 3, and the flow-based Z-Image Turbo [38] in Figure 1. Across all three, our method applies the requested edit while preserving the layout, identity, and background of the input. The baselines instead exhibit two recurring failure modes. Many under-edit: on SD, iCD, NPI, NTI, SwiftEdit, and InfEdit frequently produce little visible change (e.g., *surfboards→flowers*, *smoke→fire*). Those that do edit tend to over-edit: ChordEdit distorts the surrounding scene, ReNoise and TurboEdit sacrifice pose, background, and identity, and the flow baselines drift out of the input domain. In

contrast, our method produces a consistent, visible edit in every case; further per-backbone examples are in Appendix D.2.

Quantitative evaluation. We group metrics into four aspects, editability, preservation, quality, and efficiency, and summarize overall standing with the mean rank across metrics (Avg Rank); per-family results are in Table 2.

Editability is measured by the masked CLIP (CLIP ViT-L/14) similarities (CLIP_{wh}, CLIP_{ed}) and the directional dCLIP score (Eq. 20, additionally measured with OpenCLIP ViT-bigG/14 in Appendix D.4), reflecting how well the edited image follows the target instruction. Our method achieves the top dCLIP in every block and the top CLIP_{wh} in the SDXL and flow blocks, remaining a close second in the SD block. The dCLIP values also quantify the under-editing seen qualitatively in Figures 1, 3, and 4: many SD baselines apply very weak edits (e.g., 0.047 for iCD, 0.050 for NTI, 0.081 for SwiftEdit, against our 0.155), consistent with the figures where these methods leave the source nearly unchanged.

Preservation is assessed with Structure Distance, LPIPS (SqueezeNet), SSIM, and PSNR, capturing fidelity to the input x_0 in non-edited regions. Our method attains the best LPIPS in the flow block and stays competitive across all families. Where a baseline edges ahead on preservation, it does so at far lower edit strength (GNRi on SDXL: dCLIP 0.038) or higher inference cost (UniEdit-Flow on flow), reflecting a conservative operating point rather than a better trade-off; consistently, Structure Distance favours conservative editors, and its per-block winners (NTI, GNRi, UniEdit-Flow) edit near least. Sweeping the guidance scale ω traces a full quality–preservation frontier that dominates the baselines' single operating points (Appendix D.2).

Quality is measured by Image Reward, a learned proxy for human preference. Our method attains the best Image Reward in every block. Notably, several under-editing baselines score at or below zero (e.g., iCD, NTI on SD; GNRi on SDXL), confirming that producing little change is penalised as low-quality rather than rewarded as faithful preservation.

Efficiency. Our method yields the lowest or near-lowest inference time in each block, several times faster than the competing flow editors. It attains the best Avg Rank on the SDXL and flow backbones; on SD it ranks a close third (3.44 vs. 3.33) behind G&R and InfEdit, which are 6–30+$\times$ slower. Times are inference only: ours, iCD and SwiftEdit train once up front, the rest do not (Appendix A).

Human evaluation. A pairwise study on Z-Image-Turbo against the four flow editors (Appendix B, Table 3) confirms the metrics: ours is preferred on overall quality (76.3%) and preservation (78.8%) and at near parity on prompt match (58.5%), i.e. it applies the same edits while preserving more of the input.

6 Conclusion

We introduced a simple, universal framework for text-guided editing with accelerated backbones: inversion and generation form a single differentiable graph, and we optimize a task-level objective over the full trajectory rather than per-step surrogates. Fine-tuning two lightweight LoRA adapters (forward and backward) jointly enforces faithful reconstruction, prompt-consistent change, and a well-posed terminal latent. Across consistency-distilled, generator-based, and flow-based backbones, it attains the strongest edit alignment in each family, with preservation that matches or exceeds prior editors at matched edit strength, efficient inference, and no backbone-specific inversion or test-time optimization. Future work includes instruction-style edits, higher resolutions, and video editing.

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References

[1] Tim Brooks, Aleksander Holynski, and Alexei A. Efros. Instructpix2pix: Learning to follow image editing instructions. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2023, Vancouver, BC, Canada, June 17-24, 2023*, pages 18392–18402. IEEE, 2023. doi: 10.1109/CVPR52729.2023.01764. URL https://doi.org/10.1109/CVPR52729.2023.01764.

[2] Mingdeng Cao, Xintao Wang, Zhongang Qi, Ying Shan, Xiaohu Qie, and Yinqiang Zheng. Masactrl: Tuning-free mutual self-attention control for consistent image synthesis and editing. In *IEEE/CVF International Conference on Computer Vision, ICCV 2023, Paris, France, October 1-6, 2023*, pages 22503–22513. IEEE, 2023. doi: 10.1109/ICCV51070.2023.02062. URL https://doi.org/10.1109/ICCV51070.2023.02062.

[3] Kevin Clark, Paul Vicol, Kevin Swersky, and David J. Fleet. Directly fine-tuning diffusion models on differentiable rewards. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net, 2024. URL https://openreview.net/forum?id=1vmSEVL19f.

[4] Yasong Dai, Zeeshan Hayder, David Ahmedt-Aristizabal, and Hongdong Li. Bifm: Bidirectional flow matching for few-step image editing and generation. *CoRR*, abs/2603.24942, 2026. doi: 10.48550/ARXIV.2603.24942. URL https://doi.org/10.48550/arXiv.2603.24942.

[5] Trung Tuan Dao, Thuan Hoang Nguyen, Thanh Le, Duc Hong Vu, Khoi Nguyen, Cuong Pham, and Anh Tuan Tran. Swiftbrush V2: make your one-step diffusion model better than its teacher. In Ales Leonardis, Elisa Ricci, Stefan Roth, Olga Russakovsky, Torsten Sattler, and Gül Varol, editors, *Computer Vision - ECCV 2024 - 18th European Conference, Milan, Italy, September 29-October 4, 2024, Proceedings, Part LXXXII*, volume 15140 of *Lecture Notes in Computer Science*, pages 176–192. Springer, 2024. doi: 10.1007/978-3-031-73007-8_11. URL https://doi.org/10.1007/978-3-031-73007-8_11.

[6] Yingying Deng, Xiangyu He, Changwang Mei, Peisong Wang, and Fan Tang. Fireflow: Fast inversion of rectified flow for image semantic editing. In Aarti Singh, Maryam Fazel, Daniel Hsu, Simon Lacoste-Julien, Felix Berkenkamp, Tegan Maharaj, Kiri Wagstaff, and Jerry Zhu, editors, *Forty-second International Conference on Machine Learning, ICML 2025, Vancouver, BC, Canada, July 13-19, 2025*, volume 267 of *Proceedings of Machine Learning Research*. PMLR / OpenReview.net, 2025. URL https://proceedings.mlr.press/v267/deng25c.html.

[7] Gilad Deutch, Rinon Gal, Daniel Garibi, Or Patashnik, and Daniel Cohen-Or. Turboedit: Text-based image editing using few-step diffusion models. In Takeo Igarashi,

Ariel Shamir, and Hao (Richard) Zhang, editors, *SIGGRAPH Asia 2024 Conference Papers, SA 2024, Tokyo, Japan, December 3-6, 2024*, pages 41:1–41:12. ACM, 2024. doi: 10.1145/3680528.3687612. URL https://doi.org/10.1145/3680528.3687612.

[8] Patrick Esser, Sumith Kulal, Andreas Blattmann, Rahim Entezari, Jonas Müller, Harry Saini, Yam Levi, Dominik Lorenz, Axel Sauer, Frederic Boesel, Dustin Podell, Tim Dockhorn, Zion English, and Robin Rombach. Scaling rectified flow transformers for high-resolution image synthesis. In Ruslan Salakhutdinov, Zico Kolter, Katherine A. Heller, Adrian Weller, Nuria Oliver, Jonathan Scarlett, and Felix Berkenkamp, editors, *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024*, volume 235 of *Proceedings of Machine Learning Research*, pages 12606–12633. PMLR / OpenReview.net, 2024. URL https://proceedings.mlr.press/v235/esser24a.html.

[9] Rinon Gal, Or Patashnik, Haggai Maron, Amit H. Bermano, Gal Chechik, and Daniel Cohen-Or. Stylegan-nada: Clip-guided domain adaptation of image generators. *ACM Trans. Graph.*, 41(4):141:1–141:13, 2022. doi: 10.1145/3528223.3530164. URL https://doi.org/10.1145/3528223.3530164.

[10] Daniel Garibi, Or Patashnik, Andrey Voynov, Hadar Averbuch-Elor, and Daniel Cohen-Or. Renoise: Real image inversion through iterative noising. In Ales Leonardis, Elisa Ricci, Stefan Roth, Olga Russakovsky, Torsten Sattler, and Gül Varol, editors, *Computer Vision - ECCV 2024 - 18th European Conference, Milan, Italy, September 29-October 4, 2024, Proceedings, Part XIV*, volume 15072 of *Lecture Notes in Computer Science*, pages 395–413. Springer, 2024. doi: 10.1007/978-3-031-72630-9_23. URL https://doi.org/10.1007/978-3-031-72630-9_23.

[11] Yiming Gong, Zhen Zhu, and Minjia Zhang. Instantedit: Text-guided few-step image editing with piecewise rectified flow. In *IEEE/CVF International Conference on Computer Vision, ICCV 2025, Honolulu, HI, USA, October 19-25, 2025*, pages 16808–16817. IEEE, 2025. doi: 10.1109/ICCV51701.2025.01561. URL https://doi.org/10.1109/ICCV51701.2025.01561.

[12] Xiefan Guo, Miaomiao Cui, Liefeng Bo, and Di Huang. Shortft: Diffusion model alignment via shortcut-based fine-tuning. In *IEEE/CVF International Conference on Computer Vision, ICCV 2025, Honolulu, HI, USA, October 19-25, 2025*, pages 678–687. IEEE, 2025. doi: 10.1109/ICCV51701.2025.00071. URL https://doi.org/10.1109/ICCV51701.2025.00071.

[13] Amir Hertz, Kfir Aberman, and Daniel Cohen-Or. Delta denoising score. In *IEEE/CVF International Conference on Computer Vision, ICCV 2023, Paris, France, October 1-6, 2023*, pages 2328–2337. IEEE, 2023. doi: 10.1109/ICCV51070.2023.00221. URL https://doi.org/10.1109/ICCV51070.2023.00221.

[14] Amir Hertz, Ron Mokady, Jay Tenenbaum, Kfir Aberman, Yael Pritch, and Daniel Cohen-Or. Prompt-to-prompt image editing with cross-attention control. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023. URL https://openreview.net/forum?id=_CDixzkzeyb.

[15] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In Hugo Larochelle, Marc'Aurelio Ranzato, Raia Hadsell, Maria-Florina Balcan, and Hsuan-Tien Lin, editors, *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*, 2020. URL https://proceedings.neurips.cc/paper/2020/hash/4c5bcfec8584af0d967f1ab10179ca4b-Abstract.html.

[16] Mude Hui, Siwei Yang, Bingchen Zhao, Yichun Shi, Heng Wang, Peng Wang, Cihang Xie, and Yuyin Zhou. Hq-edit: A high-quality dataset for instruction-based image editing. In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net, 2025. URL https://openreview.net/forum?id=mZptYYttFj.

[17] Guanlong Jiao, Biqing Huang, Kuan-Chieh Wang, and Renjie Liao. Uniedit-flow: Unleashing inversion and editing in the era of flow models, 2025. URL https://arxiv.org/abs/2504.13109.

[18] Xuan Ju, Ailing Zeng, Yuxuan Bian, Shaoteng Liu, and Qiang Xu. Pnp inversion: Boosting diffusion-based editing with 3 lines of code. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net, 2024. URL https://openreview.net/forum?id=FoMZ4ljhVw.

[19] Gwanghyun Kim, Taesung Kwon, and Jong Chul Ye. Diffusionclip: Text-guided diffusion models for robust image manipulation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2022, New Orleans, LA, USA, June 18-24, 2022*, pages 2416–2425. IEEE, 2022. doi: 10.1109/CVPR52688.2022.00246. URL https://doi.org/10.1109/CVPR52688.2022.00246.

[20] Vladimir Kulikov, Matan Kleiner, Inbar Huberman-Spiegelglas, and Tomer Michaeli. Flowedit: Inversion-free text-based editing using pre-trained flow models. In *IEEE/CVF International Conference on Computer Vision, ICCV 2025, Honolulu, HI, USA, October 19-25, 2025*, pages 19721–19730. IEEE, 2025. doi: 10.1109/ICCV51701.2025.01834. URL https://doi.org/10.1109/ICCV51701.2025.01834.

[21] Black Forest Labs, Stephen Batifol, Andreas Blattmann, Frederic Boesel, Saksham Consul, Cyril Diagne, Tim Dockhorn, Jack English, Zion English, Patrick Esser, Sumith Kulal, Kyle Lacey, Yam Levi, Cheng Li, Dominik Lorenz, Jonas Müller, Dustin Podell, Robin Rombach, Harry Saini, Axel Sauer, and Luke Smith. FLUX.1 kontext: Flow matching for in-context image generation and editing in latent space. *CoRR*, abs/2506.15742, 2025. doi: 10.48550/ARXIV.2506.15742. URL https://doi.org/10.48550/arXiv.2506.15742.

[22] Tsung-Yi Lin, Michael Maire, Serge J. Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. Microsoft COCO: common objects in context. In David J. Fleet, Tomás Pajdla, Bernt Schiele, and Tinne Tuytelaars, editors, *Computer Vision - ECCV 2014 - 13th European Conference, Zurich, Switzerland,*

September 6-12, 2014, Proceedings, Part V, volume 8693 of *Lecture Notes in Computer Science*, pages 740–755. Springer, 2014. doi: 10.1007/978-3-319-10602-1_48. URL https://doi.org/10.1007/978-3-319-10602-1_48.

[23] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023. URL https://openreview.net/forum?id=PqvMRDCJT9t.

[24] Xingchao Liu, Chengyue Gong, and Qiang Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023. URL https://openreview.net/forum?id=XVjTT1nw5z.

[25] Liangsi Lu, Xuhang Chen, Minzhe Guo, Shichu Li, Jingchao Wang, and Yang Shi. Chordedit: One-step low-energy transport for image editing. *arXiv preprint arXiv:2602.19083*, 2026.

[26] Simian Luo, Yiqin Tan, Longbo Huang, Jian Li, and Hang Zhao. Latent consistency models: Synthesizing high-resolution images with few-step inference. *CoRR*, abs/2310.04378, 2023. doi: 10.48550/ARXIV.2310.04378. URL https://doi.org/10.48550/arXiv.2310.04378.

[27] Daiki Miyake, Akihiro Iohara, Yu Saito, and Toshiyuki Tanaka. Negative-prompt inversion: Fast image inversion for editing with text-guided diffusion models. In *IEEE/CVF Winter Conference on Applications of Computer Vision, WACV 2025, Tucson, AZ, USA, February 26 - March 6, 2025*, pages 2063–2072. IEEE, 2025. doi: 10.1109/WACV61041.2025.00207. URL https://doi.org/10.1109/WACV61041.2025.00207.

[28] Ron Mokady, Amir Hertz, Kfir Aberman, Yael Pritch, and Daniel Cohen-Or. Null-text inversion for editing real images using guided diffusion models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2023, Vancouver, BC, Canada, June 17-24, 2023*, pages 6038–6047. IEEE, 2023. doi: 10.1109/CVPR52729.2023.00585. URL https://doi.org/10.1109/CVPR52729.2023.00585.

[29] Trong-Tung Nguyen, Quang Nguyen, Khoi Nguyen, Anh Tuan Tran, and Cuong Pham. Swiftedit: Lightning fast text-guided image editing via one-step diffusion. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2025, Nashville, TN, USA, June 11-15, 2025*, pages 21492–21501. Computer Vision Foundation / IEEE, 2025. doi: 10.1109/CVPR52734.2025.02002. URL https://openaccess.thecvf.com/content/CVPR2025/html/Nguyen_SwiftEdit_Lightning_Fast_Text-Guided_Image_Editing_via_One-Step_Diffusion_CVPR_2025_paper.html.

[30] Yuxi Ren, Xin Xia, Yanzuo Lu, Jiacheng Zhang, Jie Wu, Pan Xie, Xing Wang, and Xuefeng Xiao. Hyper-sd: Trajectory segmented consistency model for efficient image synthesis. In Amir Globersons, Lester Mackey, Danielle Belgrave, Angela Fan, Ulrich Paquet, Jakub M. Tomczak, and Cheng Zhang, editors, *Advances in Neural Information Processing Systems 37: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*, 2024.

URL http://papers.nips.cc/paper_files/paper/2024/hash/d4e1c24ac41ff0b82ca1b171731f0b23-Abstract-Conference.html.

[31] Dvir Samuel, Barak Meiri, Haggai Maron, Yoad Tewel, Nir Darshan, Shai Avidan, Gal Chechik, and Rami Ben-Ari. Lightning-fast image inversion and editing for text-to-image diffusion models. In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net, 2025. URL https://openreview.net/forum?id=t9l63huPRt.

[32] Axel Sauer, Frederic Boesel, Tim Dockhorn, Andreas Blattmann, Patrick Esser, and Robin Rombach. Fast high-resolution image synthesis with latent adversarial diffusion distillation. In Takeo Igarashi, Ariel Shamir, and Hao (Richard) Zhang, editors, *SIGGRAPH Asia 2024 Conference Papers, SA 2024, Tokyo, Japan, December 3-6, 2024*, pages 106:1–106:11. ACM, 2024. doi: 10.1145/3680528.3687625. URL https://doi.org/10.1145/3680528.3687625.

[33] Axel Sauer, Dominik Lorenz, Andreas Blattmann, and Robin Rombach. Adversarial diffusion distillation. In Ales Leonardis, Elisa Ricci, Stefan Roth, Olga Russakovsky, Torsten Sattler, and Gül Varol, editors, *Computer Vision - ECCV 2024 - 18th European Conference, Milan, Italy, September 29-October 4, 2024, Proceedings, Part LXXXVI*, volume 15144 of *Lecture Notes in Computer Science*, pages 87–103. Springer, 2024. doi: 10.1007/978-3-031-73016-0_6. URL https://doi.org/10.1007/978-3-031-73016-0_6.

[34] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *9th International Conference on Learning Representations, ICLR 2021, Virtual Event, Austria, May 3-7, 2021*. OpenReview.net, 2021. URL https://openreview.net/forum?id=St1giarCHLP.

[35] Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *9th International Conference on Learning Representations, ICLR 2021, Virtual Event, Austria, May 3-7, 2021*. OpenReview.net, 2021. URL https://openreview.net/forum?id=PxTIG12RRHS.

[36] Yang Song, Prafulla Dhariwal, Mark Chen, and Ilya Sutskever. Consistency models. In Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett, editors, *International Conference on Machine Learning, ICML 2023, 23-29 July 2023, Honolulu, Hawaii, USA*, volume 202 of *Proceedings of Machine Learning Research*, pages 32211–32252. PMLR, 2023. URL https://proceedings.mlr.press/v202/song23a.html.

[37] Nikita Starodubcev, Mikhail Khoroshikh, Artem Babenko, and Dmitry Baranchuk. Invertible consistency distillation for text-guided image editing in around 7 steps. In Amir Globersons, Lester Mackey, Danielle Belgrave, Angela Fan, Ulrich Paquet, Jakub M. Tomczak, and Cheng Zhang, editors, *Advances in Neural Information Processing Systems 37: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*, 2024. URL http://papers.nips.cc/paper_files/paper/2024/hash/1704fe7aaff33a54802b83a016050ab8-Abstract-Conference.html.

[38] Z-Image Team, Huanqia Cai, Sihan Cao, Ruoyi Du, Peng Gao, Steven Hoi, Zhaohui Hou, Shijie Huang, Dengyang Jiang, Xin Jin, Liangchen Li, Zhen Li, Zhong-Yu Li, David Liu, Dongyang Liu, Junhan Shi, Qilong Wu, Feng Yu, Chi Zhang, Shifeng Zhang, and Shilin Zhou. Z-image: An efficient image generation foundation model with single-stream diffusion transformer. *arXiv preprint arXiv:2511.22699*, 2025. URL https://arxiv.org/abs/2511.22699.

[39] Feng Tian, Yixuan Li, Yichao Yan, Shanyan Guan, Yanhao Ge, and Xiaokang Yang. Postedit: Posterior sampling for efficient zero-shot image editing. In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net, 2025. URL https://openreview.net/forum?id=J8YWCBPgx7.

[40] Vadim Titov, Madina Khalmatova, Alexandra Ivanova, Dmitry P. Vetrov, and Aibek Alanov. Guide-and-rescale: Self-guidance mechanism for effective tuning-free real image editing. In Ales Leonardis, Elisa Ricci, Stefan Roth, Olga Russakovsky, Torsten Sattler, and Gül Varol, editors, *Computer Vision - ECCV 2024 - 18th European Conference, Milan, Italy, September 29-October 4, 2024, Proceedings, Part LXXI*, volume 15129 of *Lecture Notes in Computer Science*, pages 235–251. Springer, 2024. doi: 10.1007/978-3-031-73209-6_14. URL https://doi.org/10.1007/978-3-031-73209-6_14.

[41] Jiangshan Wang, Junfu Pu, Zhongang Qi, Jiayi Guo, Yue Ma, Nisha Huang, Yuxin Chen, Xiu Li, and Ying Shan. Taming rectified flow for inversion and editing. In Aarti Singh, Maryam Fazel, Daniel Hsu, Simon Lacoste-Julien, Felix Berkenkamp, Tegan Maharaj, Kiri Wagstaff, and Jerry Zhu, editors, *Forty-second International Conference on Machine Learning, ICML 2025, Vancouver, BC, Canada, July 13-19, 2025*, volume 267 of *Proceedings of Machine Learning Research*. PMLR / OpenReview.net, 2025. URL https://proceedings.mlr.press/v267/wang25ce.html.

[42] Tao Wu, Senmao Li, Yaxing Wang, Shiqi Yang, Kai Wang, and Joost van de Weijer. Training-free image inversion for one-step diffusion models. *Pattern Recognit.*, 180: 114063, 2026. doi: 10.1016/J.PATCOG.2026.114063. URL https://doi.org/10.1016/j.patcog.2026.114063.

[43] Zongze Wu, Nicholas I. Kolkin, Jonathan Brandt, Richard Zhang, and Eli Shechtman. Turboedit: Instant text-based image editing. In Ales Leonardis, Elisa Ricci, Stefan Roth, Olga Russakovsky, Torsten Sattler, and Gül Varol, editors, *Computer Vision - ECCV 2024 - 18th European Conference, Milan, Italy, September 29-October 4, 2024, Proceedings, Part LXXX*, volume 15138 of *Lecture Notes in Computer Science*, pages 365–381. Springer, 2024. doi: 10.1007/978-3-031-72989-8_21. URL https://doi.org/10.1007/978-3-031-72989-8_21.

[44] Jiazheng Xu, Xiao Liu, Yuchen Wu, Yuxuan Tong, Qinkai Li, Ming Ding, Jie Tang, and Yuxiao Dong. Imagereward: Learning and evaluating human preferences for text-to-image generation. In Alice Oh, Tristan Naumann, Amir Globerson, Kate Saenko, Moritz Hardt, and Sergey Levine, editors, *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10 - 16, 2023*, 2023.

URL http://papers.nips.cc/paper_files/paper/2023/hash/33646ef0ed554145eab65f6250fab0c9-Abstract-Conference.html.

[45] Sihan Xu, Yidong Huang, Jiayi Pan, Ziqiao Ma, and Joyce Chai. Inversion-free image editing with language-guided diffusion models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2024, Seattle, WA, USA, June 16-22, 2024*, pages 9454–9461. IEEE, 2024. doi: 10.1109/CVPR52733.2024.00903. URL https://doi.org/10.1109/CVPR52733.2024.00903.

[46] Tianwei Yin, Michaël Gharbi, Richard Zhang, Eli Shechtman, Frédo Durand, William T. Freeman, and Taesung Park. One-step diffusion with distribution matching distillation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2024, Seattle, WA, USA, June 16-22, 2024*, pages 6613–6623. IEEE, 2024. doi: 10.1109/CVPR52733.2024.00632. URL https://doi.org/10.1109/CVPR52733.2024.00632.

[47] Jing Zuo, Luoping Cui, Chuang Zhu, and Yonggang Qi. Precise diffusion inversion: Towards novel samples and few-step models. In Danielle Belgrave, Cheng Zhang, Laura N. Montoya, Hsuan-Tien Lin, Razvan Pascanu, Piotr Koniusz, Marzyeh Ghassemi, Nancy Chen, Iván Vladimir Meza Ruíz, and Arturo Loaiza-Bonilla, editors, *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2025, NeurIPS 2025, San Diego, CA, USA, December 2-7, 2025 / Mexico City, Mexico, November 30 - December 5, 2025*, 2025. URL http://papers.nips.cc/paper_files/paper/2025/hash/2e4bc9f6e31aa27861299940aa5242ad-Abstract-Conference.html.

Appendix

A Fine-tune setup

All backbone models were initialized from publicly available checkpoints. We used the default inference configuration for each accelerated model, namely 4 sampling steps for SD1.5 and SDXL-Turbo, and 8 sampling steps for Z-Image Turbo. For each backbone, we fine-tuned two LoRA adapters of identical architecture, corresponding to the forward model F_ϕ and the backward model G_θ. The LoRA rank was set to 32 for SD1.5 and 64 for SDXL-Turbo and Z-Image Turbo, with adapters applied to the attention projections `[to_k, to_v, to_q, to_out.0]`. All loss terms used a weight of 1.0, except the KL term, which was weighted by 0.5 for SD1.5 and 0.1 for SDXL-Turbo and Z-Image Turbo; the low-frequency term $\mathcal{L}_{\text{LF}}$ used a pooling size $k = 8$. Classifier-free guidance was disabled during fine-tuning. The overall scheme of our pipeline is shown in Figure 2. All models were optimized with AdamW at a learning rate of 4×10^{-4} under a cosine schedule, on 4 NVIDIA H100 GPUs. We used a batch size of 16 for SD1.5 and SDXL-Turbo, and 4 for Z-Image Turbo. Fine-tuning was run until convergence, which took 300 iterations for SD1.5 and 600 for SDXL-Turbo and Z-Image Turbo; we use the final checkpoint directly. The full recipe costs $\approx$ 15 H100 GPU-hours once per backbone (300 iterations for SD1.5, 600 for SDXL-Turbo and Z-Image Turbo).

Loss-weight sweeps. Table 10 retrains the model at each weight setting and evaluates all runs at the Table 2 operating point. No setting improves on the default: every non-default weight lands on or strictly inside the quality–preservation frontier that the default model already traces by sweeping ω. Raising λ_{LF} strengthens edits at a preservation cost, but no better than raising ω instead (λ_{LF}=2: dCLIP 0.171 at LPIPS 0.264, vs. the default at $\omega\approx$3.4: 0.171 at 0.224); the model trains stably across the full 8$\times$ range. The KL weight has a narrower useful band: λ_{KL}=0 weakens edits (dCLIP 0.154 $\rightarrow$ 0.135), λ_{KL}=0.05 is strictly dominated by the default at ω=3 (same dCLIP, LPIPS 0.240 vs. 0.212), and λ_{KL}=0.5 collapses training on this backbone (IR -2.27): the prior overwhelms reconstruction and z_T stops carrying the source. The useful KL band is backbone-dependent: the sweep above is on Z-Image Turbo, whereas the SD1.5 recipe trains stably at its default λ_{KL}=0.5. The per-backbone defaults are $(\lambda_{\text{LF}}, \lambda_{\text{KL}}) = (1, 0.5)$ for SD1.5 and $(1, 0.1)$ for SDXL-Turbo and Z-Image Turbo, used everywhere else in the paper.

Parameter cost and comparison to fine-tuning-based editors. Our method trains only two LoRA adapters (forward inversion, backward denoising) on a frozen backbone. Counting directly from the checkpoints, this is 12.75M, 185.79M, and 133.69M trainable parameters for SD1.5, SDXL-Turbo, and Z-Image Turbo respectively (Table 5), split evenly across the two adapters and amounting to 1.5–7% of the frozen base. This is markedly lighter than prior fine-tuning-based fast editors in procedure, not just count: iCD [37] obtains its inversion through consistency distillation, and SwiftEdit [29] fully trains a dedicated $\sim$921M one-step inversion network in two stages on top of an $\sim$866M generator. We add no separate inversion network and no distillation: the same single-stage recipe (300–600 iterations) applies across U-Net and DiT backbones.

A.1 Relation to SwiftEdit

Our pipeline shares its outer shape with SwiftEdit [29]: invert to a latent, regenerate under the target prompt, regularize the inverted latent. It differs in every component that shape is built from.

What is trained. SwiftEdit trains a dedicated one-step inversion network of the same scale as its ∼866M generator, in two stages (synthetic images from SwiftBrushv2, then real ones), with an IP-Adapter-style conditioning branch. We train no new network: two LoRA adapters (12.75–185.79M, 1.5–7% of the frozen base) on a single backbone, in one stage of 300–600 iterations on fewer than 10K triplets.

One-sided vs. two-sided. SwiftEdit trains only the inversion side; the generator enters the loop as a fixed one-step decoder, and editing is steered at inference by a self-guided mask and attention rescaling. We adapt both directions and calibrate them to each other, and our inference uses no attention control and no masks. Figure 12 isolates this: training the forward adapter alone is unstable and settles at lower dCLIP and ImageReward, because the fixed denoiser is not adapted to the learned terminal latent.

What the latent regularizer constrains. Both methods regularize the inverted latent toward the noise prior, but channel-space statistics alone are permutation-blind: a standardized image latent attains channel-KL of 0.000 while carrying 16.9× the white-noise low-frequency energy (Fig. 5). Our low-frequency hinge (Eq. 18) closes this blind spot, and Fig. 8 shows the statistic it penalizes is causally harmful: injecting low-frequency structure at fixed channel-KL degrades editing monotonically.

Scope. SwiftEdit is re-instantiable on other one-step generators (DMD2, InstaFlow, SwiftBrush), but each re-initializes and retrains the inversion network through both stages, and all demonstrated backbones remain U-Net (SD1.5/SD2.1). The same single-stage recipe here transfers unchanged across three distillation paradigms and to a DiT flow backbone.

B Human preference study

Protocol. We use the first 50 images of PIE-Bench [18] and, for each, edit it with our method and with every baseline (RF-Edit [41], FlowEdit [20], FireFlow [6], UniEdit-Flow [17]). A rater sees the source image and prompt, the edit prompt, and the two edited results (ours and a baseline) in randomized order, and answers which they prefer overall, which better matches the edit prompt, and which better preserves the source; a tie is allowed. This yields 50 comparisons per baseline (200 total), scored as win $+\frac{1}{2}$tie (50% = parity). Each comparison is judged independently by two annotators, who answer all three questions (overall preference, prompt match, source preservation). For any question on which their answers differ, including cases where one chooses a side and the other a tie, an expert rater inspects that comparison and provides the rating, which is taken as final. In total four people participated: two fixed annotators across all 200 comparisons and two expert adjudicators. Prior to adjudication, the two annotators agreed on 97.5% of question-level judgments.

Results. Our method is preferred over every baseline on overall quality and preservation, with wins far outnumbering losses (128/23 and 138/23 of 200). Prompt match is near

parity, dominated by ties, so our edits follow the prompt about as faithfully as the baselines; the decisive gap is preservation, where baselines reach comparable edit fidelity only by altering the source. Parity on this question is expected rather than in tension with the dCLIP gaps in Table 2: the prompt-match judgment is effectively binary: raters decide whether the requested edit is present, not how strongly it is applied, so it saturates once any adequate edit is made. The automatic metrics show the same compression: under the official protocol, CLIP_{ed} spans only 20.7–23.3 across the methods of Table 2, against a no-edit floor of 20.1 (scoring the unedited source images), whereas the graded dCLIP separates the same methods several-fold within each block. This mirrors Table 3: we match competing flow editors on edit strength while preserving more of the input.

C Ablations

To understand what drives our method, we provide a detailed analysis of each component of our optimization strategy: the forward and backward adapters (Fig. 12) and the individual loss terms (Table 1). Throughout the appendices, LPIPS_{wh} (VGG-16) denotes whole-image VGG-16 LPIPS, and LPIPS-pres/PSNR-pres denote the same metrics computed outside the edit mask by our internal scorer. Both differ from the official background-masked SqueezeNet LPIPS and masked PSNR of Table 2 and are not comparable to them in absolute value.

C.1 Single adapter ablation

We analyze the contribution of each adapter by fine-tuning only the forward adapter F_ϕ, only the backward adapter G_θ, and both jointly (our full method), and track LPIPS, dCLIP, and ImageReward on PIE-Bench over training (Fig. 12). Neither single-adapter variant is sufficient. Training only G_θ leaves reconstruction poor: LPIPS plateaus early and never approaches the joint model, since the backward pass cannot compensate for an uncalibrated inversion distribution. Training only F_ϕ reaches better LPIPS than the backward-only variant but is unstable across training and converges to a worse operating point, with lower CLIP-Edit and ImageReward, because the fixed denoiser is not adapted to the learned terminal latent. Optimizing both adapters jointly resolves both failure modes, attaining substantially lower LPIPS while remaining competitive on dCLIP and ImageReward throughout training, confirming that inversion and generation must be calibrated to each other rather than tuned in isolation.

C.2 Analysis of the learned terminal latents

Complementarity of the two priors. The priors are complementary by construction. A per-channel standardized image latent attains channel-KL of exactly 0 while its low-frequency energy is 16.9× the white-noise floor (Fig. 5); conversely, under-scaled white noise ($0.77\,\varepsilon$, the empirical channel scale of the base model's terminal latents) passes the one-sided LF hinge (0.60) but fails the KL term (0.058). On MS-COCO (n=2700) the KL term drives the median per-image channel-KL of the inverted latents from 0.0186 (trained without it) to 0.0032. Symmetrically for the LF statistic: without the hinge, the pooled low-frequency excess reaches 4.1× the white-noise floor in the worst case (median 1.73); re-inverting that worst case with the full model removes ≈35% of its low-frequency energy (4.1× → 2.6×),

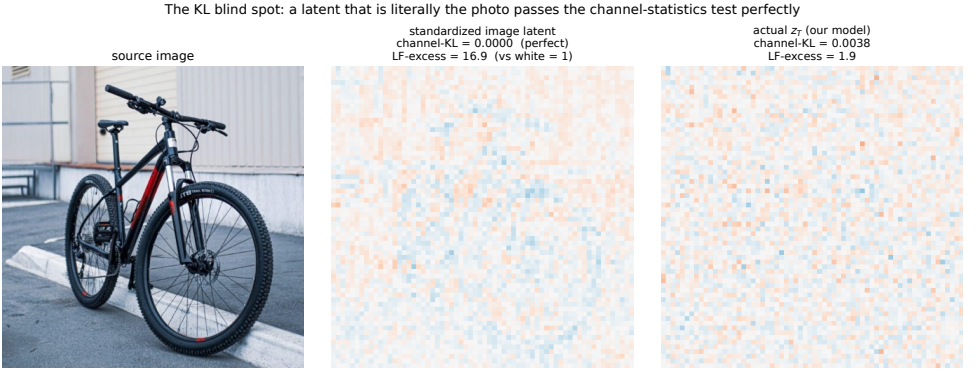


Figure 5: **The KL blind spot.** A per-channel standardized image latent (middle; the bicycle is faintly visible) passes the channel-statistics test perfectly (KL = 0.000) yet carries 16.9× the white-noise low-frequency energy; our inverted z_T (right) passes both tests.

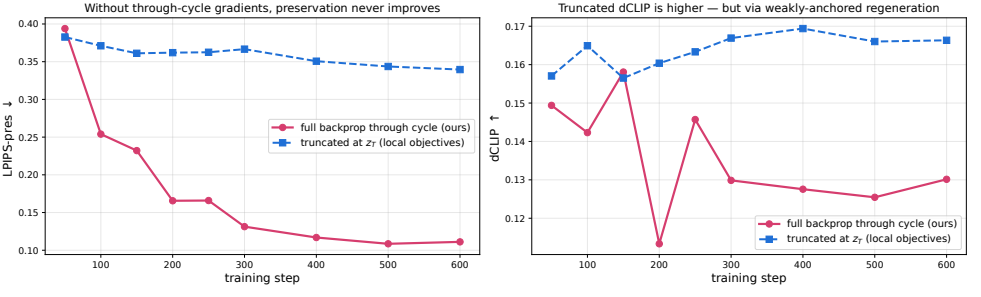


Figure 6: **Gradient-flow ablation:** same recipe with gradients detached at z_T (Z-Image Turbo, ω=2.5). Left: without through-cycle gradients, preservation never improves. Right: the truncated model's higher dCLIP reflects weakly-anchored regeneration from the caption, not editing of the input. LPIPS-pres: VGG-16 LPIPS outside the edit mask.

and the population median drops to 1.61. Increasing λ_{LF} to 2 suppresses the tail entirely (99th percentile 2.55 → 1.85).

Causal effect of LF structure. Injecting low-frequency noise into z_T at increasing strength, renormalized per channel so that channel-KL is unchanged by construction, monotonically degrades editing (Fig. 8): the statistic penalized by the LF hinge is causally harmful, consistent with the frontier ablation (Fig. 11), where the without-LF frontier is the weakest.

Composition and interpolation. Splicing two inverted latents half-and-half (30 PIE-Bench pairs) decodes each half near round-trip quality: LPIPS 0.195 vs. 0.144 for full reconstruction of the same parent images under the same decoding protocol, and 0.775 against the wrong source. Interpolating pairs of inverted latents yields monotonic morphs on 97–100% of pairs with path length 1.59× the direct distance, vs. 2.95× for raw re-noised latents (Fig. 9).

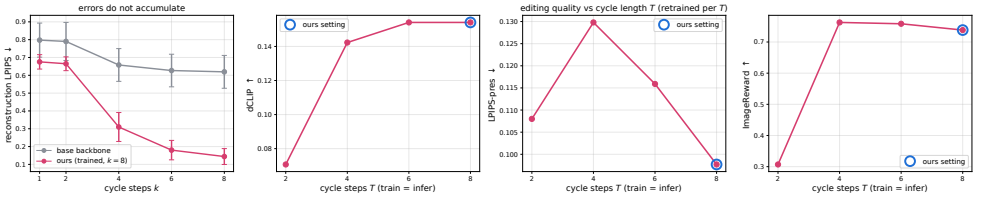


Figure 7: **Cycle length.** Left: reconstruction LPIPS (VGG-16) vs. cycle steps k (PIE-Bench, n=50); errors do not accumulate, the trained model improves with k while the base backbone stays flat. Right three panels: retraining the same recipe at $T \in \{2,4,6,8\}$ and evaluating each at its native T; editing saturates at T=6, preservation peaks at T=8 (our setting, circled).

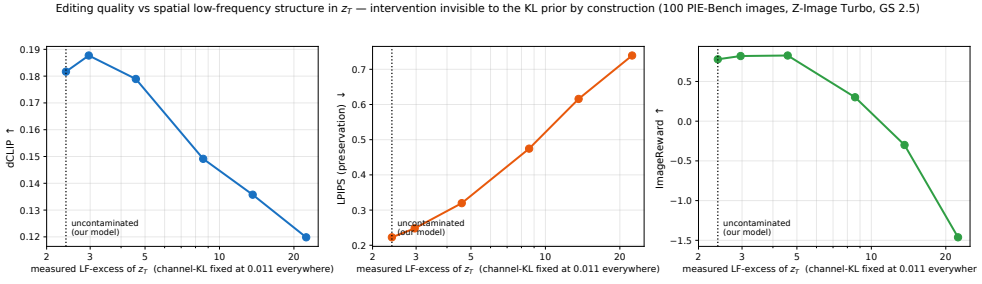


Figure 8: **Causal LF-contamination experiment** (PIE-Bench, n=100, ω=2.5). Low-frequency noise is injected into z_T at increasing strength and renormalized per channel, so channel-KL is unchanged by construction; editing quality nevertheless degrades monotonically in dCLIP, preservation LPIPS, and ImageReward. The statistic penalized by the LF hinge is causally harmful, not merely correlated. Channel-KL here is the mean per-image statistic over the 100 PIE-Bench source images (right-skewed; median 0.006) and is unchanged by the intervention by construction; the MS-COCO medians quoted in Sec. C.2 are computed identically on a different image set.

C.3 Global vs. local optimization and cycle length

Truncating the gradient at z_T. To isolate the effect of optimizing through the full cycle, we retrain the Z-Image recipe with gradients detached at the z_T boundary, so each adapter sees only its local objective. At our default operating point (guided inference of Eq. 23, ω=2.5), the truncated model collapses reconstruction fidelity: LPIPS-pres 0.338 vs. 0.098 for the full model, PSNR-pres 11.7 vs. 24.0. Its dCLIP and ImageReward nevertheless stay high (0.174/0.77): with a fixed denoiser uncalibrated to the learned inversion, the backward pass regenerates from the caption instead of reconstructing the image. The training dynamics (Fig. 6, shown under uncond CFG) follow the same pattern: the truncated model's preservation never improves over 600 steps. Global optimization is therefore load-bearing, not merely convenient.

Error does not accumulate with cycle length. Reconstruction LPIPS improves, rather than degrades, as the number of cycle steps grows (Fig. 7, left): the trained model reaches 0.14 at k=8 while the base backbone stays at $\sim$0.6 at every k. Retraining the same recipe at $T \in \{2,4,6,8\}$ and evaluating each at its native T under delta CFG (Fig. 7, right), editing saturates at T=6 (dCLIP 0.154 at both T=6 and 8) while preservation peaks at T=8 (LPIPS-

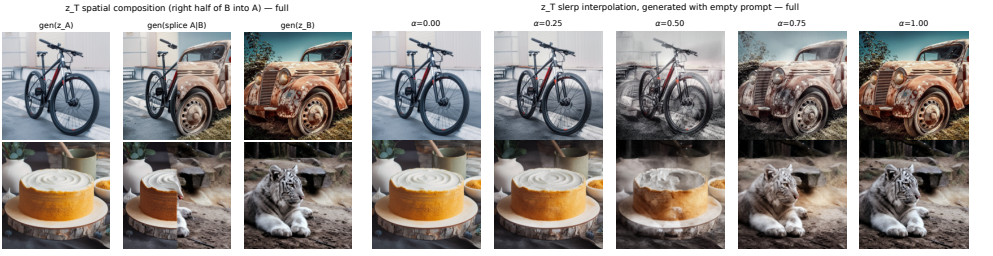


Figure 9: **The learned terminal latents compose and interpolate.** Left: splicing the right half of z_B into z_A decodes each half near round-trip quality. Right: slerp interpolation between two inverted latents (decoded with an empty prompt) yields smooth semantic morphs.

Table 5: Trainable parameters of our two adapters, counted directly from the checkpoints. The forward and backward LoRA are identical in size; "% of base" is the total trainable fraction of the frozen backbone.

Backbone	LoRA rank	Per adapter	Total trainable	Frozen base	% of base
SD1.5 (LCM)	32	6.38 M	12.75 M	∼865 M	∼1.5%
SDXL-Turbo	64	92.90 M	185.79 M	∼2.6 B	∼7%
Z-Image-Turbo	64	66.85 M	133.69 M	∼6.2B	∼2.17%

pres 0.098 vs. 0.116, PSNR-pres 24.0 vs. 22.9), our default.

C.4 Final objective

We ablate each term of the training objective on Z-Image Turbo, tracking all metrics on PIE-Bench over training (Fig. 10) together with their effect on the quality–preservation frontier (Fig. 11). *Directional CLIP is necessary for editing.* Removing the edit term entirely (*No CLIP*) drives the model toward the identity solution: it attains the lowest LPIPS (best preservation) but collapses on dCLIP and ImageReward, reconstructing the source rather than editing it. Replacing our directional objective with an absolute CLIP-score objective (*CLIPScore, no dCLIP*) fails in the same way: it produces little edit signal and tracks close to the no-edit variant on both dCLIP and CLIP Score, confirming that the edit must be driven by the source-to-target *direction* rather than absolute alignment.

The prior regularizers shift the frontier. Removing either terminal-prior term degrades the whole quality–preservation trade-off rather than a single operating point (Fig. 11): both *No KL* and *No LF* frontiers lie below and to the right of the full model across the ω sweep, with *No LF* the weakest. This indicates that constraining the terminal latent toward the backbone's noise prior, in both channel statistics (KL) and spatial whiteness (LF), is what lets the model reach strong edits without sacrificing preservation.

D Additional results

D.1 Inversion

We assess inversion through source reconstruction: we invert an image to the terminal latent $z_T = f_\phi(x_0, y_{\text{src}})$ with the forward adapter, then denoise back from z_T under the same source

prompt with the backward adapter, $\hat{x}_{\text{src}} = g_\theta(z_T, y_{\text{src}})$. No edit is applied, so a faithful method returns the input; classifier-free guidance is disabled for all methods in this setting. We compare only against methods that perform an explicit inversion, since inversion-free editors do not recover a source latent (Table 4).

Quantitatively, our adapters give the most faithful reconstruction in every backbone family, attaining the lowest MSE and LPIPS, and on the SD backbone surpassing even full-step DDIM inversion despite operating in the few-step regime (Table 4). We show representative reconstructions per backbone in Figures 17–19: Z-Image Turbo against the flow-based inversions (Fig. 17), SDXL-Turbo against the SDXL inversions (Fig. 18), and SD1.5 against DDIM, iCD, and SwiftEdit (Fig. 19). Prior accelerated inversions either lose fine detail and identity or drift in color and structure, whereas ours preserves the input closely across diverse content.

D.2 Editing

At inference our method uses only the two trained adapters, with no attention control, masks, or per-image optimization. Given an image x_0 with source prompt y_{src} and target prompt y_{trg}, we invert with the forward adapter to obtain $z_T = f_\phi(x_0, y_{\text{src}})$, then denoise from the same z_T with the guided backward adapter $\hat{g}_\theta(z_T, y_{\text{trg}}, \omega)$ to produce the edit (Sec. 4.3); the pipeline is a single inversion→generation pass. Both stages are deterministic: no noise is sampled at any point, so a given $(x_0, y_{\text{src}}, y_{\text{trg}}, \omega)$ always produces the same edit, and results are exactly reproducible. Consequently the pipeline does not offer diversity by resampling; controlled variation of the output is obtained by moving along the ω axis, which changes edit strength continuously.

Because the edit is controlled by a single guidance scale ω, sweeping ω traces an entire quality–preservation frontier rather than a single operating point. Figure 13 plots ImageReward against LPIPS as ω varies (markers labelled with ω) on all three backbones. Across every backbone our frontier lies above and to the left of the baseline operating points: for any preservation level a baseline reaches, our method attains higher edit quality at equal or lower LPIPS. Since each baseline is a single fixed point dominated by our curve, the trade-off reported at our chosen ω in Table 2 reflects the whole operating range; we use $\omega = 2.0$ for SD1.5, $\omega = 3.0$ for SDXL-Turbo, and $\omega = 2.5$ for Z-Image Turbo.

We provide further qualitative results on each backbone family in Figures 14–16: Z-Image Turbo against the flow-based editors (Fig. 14), SDXL-Turbo against the SDXL editors (Fig. 15), and SD1.5 against the SD editors (Fig. 16). Across diverse edit types, including object replacement, material and style change, attribute edits, and scene additions, our method applies the requested change while preserving layout, identity, and background, whereas the baselines tend either to under-edit or to drift from the source. Baselines are run with official code and default settings; for GNRi we additionally swept the edit CFG over [1,5] and report its best setting.

D.3 Comparison with optimization-based few-step inversion

PreciseInv [47] and TFinv [42] refine the inverted latent by per-image test-time optimization (∼600 optimization steps per image), making inference 6–16× slower than our amortized inversion for PreciseInv and ≈275× slower for TFinv (Table 6). We ran both with the official code on the full 700-image PIE-Bench, selecting each baseline's CFG by a calibration

Table 6: Optimization-based few-step inversion vs. ours under the official PIE-Bench protocol (dCLIP and Image Reward from our scorer). Both baselines optimize per image at test time; ours is amortized. Time is per-image inference on the same H100 GPUs as Table 2; the baselines' per-image optimization early-stops, hence faster than the nominal step count.

	$\text{CLIP}_{\text{wh}}\uparrow$	$\text{CLIP}_{\text{ed}}\uparrow$	dCLIP↑	Struct↓	LPIPS↓	SSIM↑	PSNR↑	IR↑	Time↓ (s)
SD-based									
PreciseInv	**26.65**	**22.95**	0.129	33.0	0.132	0.765	21.9	0.520	3.4
TFinv	24.53	21.48	0.072	45.0	0.133	0.744	19.8	0.036	165
Ours	26.13	22.47	**0.155**	**22.8**	**0.071**	**0.843**	**23.6**	**0.573**	**0.6**
SDXL-based									
PreciseInv	**27.49**	**24.08**	0.151	64.6	0.213	0.735	18.2	0.736	9.4
Ours	27.17	23.31	**0.178**	**43.0**	**0.105**	**0.808**	**20.8**	**0.858**	**0.6**

sweep; Table 6 reports the official protocol together with dCLIP and ImageReward. PreciseInv attains slightly higher CLIP similarities, but our method leads on directional alignment (dCLIP), ImageReward, and every preservation metric on both backbones. This also answers whether per-image optimization upper-bounds our approach: it does not, since the objective these methods optimize at test time is reconstruction of the source, which constrains neither alignment with the edit direction nor preservation once the prompt is changed. Training through the full inversion–editing cycle optimizes exactly those quantities, which is why the amortized inverter leads on dCLIP, ImageReward, and all preservation metrics (trailing only on raw CLIP similarity) while being up to two orders of magnitude faster.

D.4 Metric circularity and held-out re-scoring

Our training objective uses two pretrained networks: CLIP ViT-L/14 (directional loss, Eq. 20) and VGG-16 (LPIPS reconstruction loss, Eq. 16). All metrics of the official protocol are fixed by the benchmark and applied identically to every method; in particular, its LPIPS uses SqueezeNet, not the VGG-16 of our loss, Structure Distance uses DINO ViT-B/8, and SSIM, PSNR, and MSE are model-free. Of the scores we add to the protocol (dCLIP, Image Reward, runtime), only dCLIP shares a model with training: its ViT-L/14 encoder. To rule out encoder-specific bias we re-score every method in Table 2 with OpenCLIP ViT-bigG/14, an encoder unused in training with a different pretraining corpus. Table 9 shows that within-block orderings are identical and relative gaps persist under the held-out encoder; Image Reward is an independently trained reward model, and the human study (Appendix B) involves no model at all.

D.5 Per-category comparison

Table 8 breaks the comparison down by the ten PIE-Bench edit categories, reporting the within-block Avg Rank (computed as in Table 2) of our method against the best accelerated competitor. Ours ranks first in 24 of 30 backbone×category cells (26 with ties; 21/30 when the full-step SD editors are also included). The hardest category is color-attribute editing on SD/SDXL: the strong recolor (dCLIP 0.26–0.28) tends to delocalize, a failure mode absent on the flow backbone (per-category preserve-LPIPS 0.22/0.34 vs. 0.15).

Table 7: Masked-inference variant on the SD backbone (same checkpoint and ω=2.0 as Table 2); official PIE-Bench protocol, dCLIP and Image Reward from our scorer. dCLIP (0.120) still matches or exceeds every SD baseline (the closest, ChordEdit, scores 0.119)

	$\text{CLIP}_{\text{wh}}\uparrow$	$\text{CLIP}_{\text{ed}}\uparrow$	dCLIP↑	Struct↓	LPIPS↓	SSIM↑	PSNR↑	IR↑
Ours	**26.13**	**22.47**	**0.155**	22.8	0.071	0.843	23.6	**0.573**
Ours (masked)	25.43	21.85	0.120	**14.3**	**0.051**	**0.864**	**25.8**	0.440

Table 8: Per-category Avg Rank vs. accelerated editors (lower is better): ours against the best competitor in each cell. Bold marks the cell winner; = marks ties.

	SD		SDXL		Flow	
Category	Ours	Best comp.	Ours	Best comp.	Ours	Best comp.
Random	**2.33**	2.44 (InfEdit)	**1.89**	2.11 (TurboEdit)	**1.33**	2.67 (UniEdit)
Change object	**2.22**	2.33 (InfEdit)	**1.67**	2.22 (TurboEdit)	**2.33**	2.67 (FlowEdit)
Add object	**1.44**	2.89 (InfEdit)	**1.56**	2.33 (TurboEdit)	**1.56**	2.67 (UniEdit)
Delete object	**2.00**	2.67 (InfEdit)	**1.78**	2.22 (GNRi)	=2.78	2.78 (FlowEdit)
Content	**1.44**	2.67 (InfEdit)	**1.78**	2.11 (TurboEdit)	**1.56**	2.67 (UniEdit)
Pose	2.44	**1.89** (InfEdit)	**1.67**	2.22 (TurboEdit)	**2.11**	2.89 (UniEdit)
Color	=2.67	2.67 (InfEdit)	2.11	**2.00** (TurboEdit)	**1.56**	2.67 (UniEdit)
Material	**2.11**	2.44 (InfEdit)	**1.67**	2.22 (TurboEdit)	**1.33**	2.22 (UniEdit)
Background	2.44	**2.22** (InfEdit)	2.22	**1.89** (TurboEdit)	**1.78**	2.67 (UniEdit)
Style	**2.22**	2.56 (InfEdit)	**1.33**	2.33 (TurboEdit)	**1.56**	1.78 (UniEdit)

D.6 Masked-inference variant

The directional CLIP objective (Eq. 20) is global, so strong edits can spill onto semantically related content. We evaluate an inference-time masked variant on the SD backbone: cross-attention maps of the edit tokens define an edit-region mask, and the output blends the edited image inside the mask with the source reconstruction outside; no additional training is required. Table 7 uses the same checkpoint and guidance scale as the "Ours" row of Table 2, so masking is the only changed variable. The mask trades edit strength for markedly better preservation: Structure Distance improves from 22.8 to 14.3 and SSIM becomes the best in the SD block, while dCLIP (0.120) still exceeds every SD baseline in Table 2.



Figure 10: **Objective ablation on Z-Image Turbo over training iterations** (PIE-Bench, ω=2.5). LPIPS$_{wh}$(VGG-16)↓, CLIP-Score↑, dCLIP↑, and ImageReward↑ for the full objective and variants that remove the LF term, the KL term, or the edit term (*No CLIP*), and that replace directional dCLIP with absolute CLIPScore. Without a directional edit term the model collapses toward the identity solution (low LPIPS, near-zero dCLIP and ImageReward); the full objective gives the best edit quality across training.

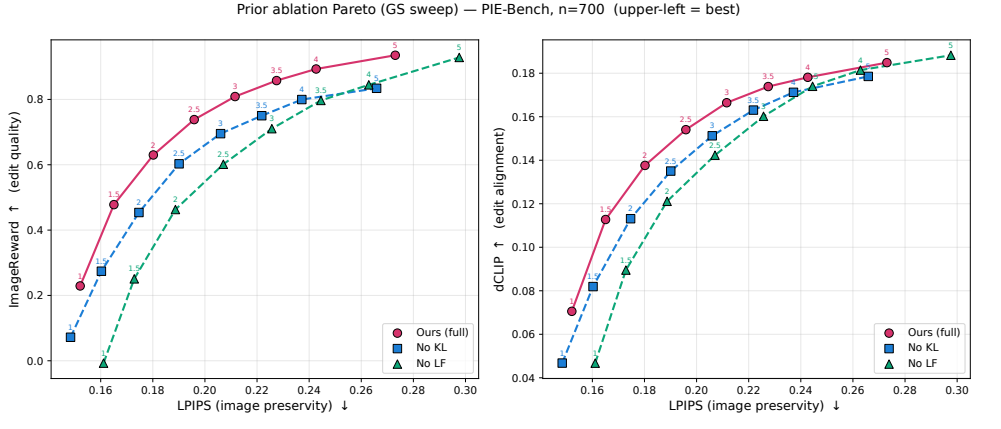


Figure 11: **Prior-regularizer ablation on Z-Image Turbo: quality–preservation frontiers** (PIE-Bench, n=700, ω sweep). ImageReward↑ (left) and dCLIP↑ (right) vs. LPIPS$_{wh}$(VGG-16)↓; upper-left is best, markers labelled with the guidance scale ω. Removing either the KL or the low-frequency (LF) term shifts the frontier below and to the right of the full model, with *No LF* the weakest.

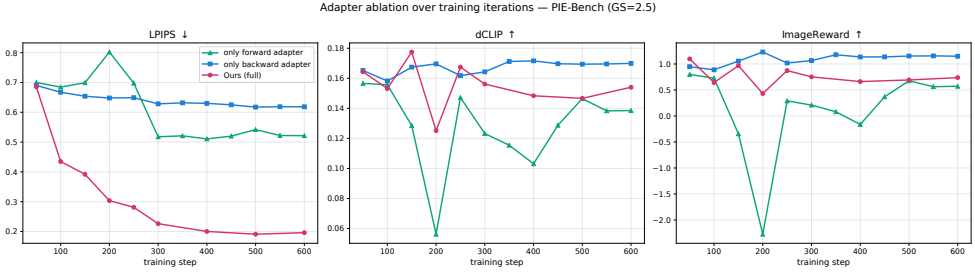


Figure 12: **Adapter ablation over training iterations** (PIE-Bench, ω=2.5). LPIPS_{wh}(VGG-16)↓, dCLIP↑, and ImageReward↑ for three training regimes: only the forward adapter F_ϕ, only the backward adapter G_θ, and both jointly (full). The backward-only model plateaus at high LPIPS; the forward-only model is unstable and settles at lower dCLIP and ImageReward. Joint training of both adapters reaches markedly lower LPIPS while remaining competitive on edit quality, showing that inversion and generation must be calibrated jointly.

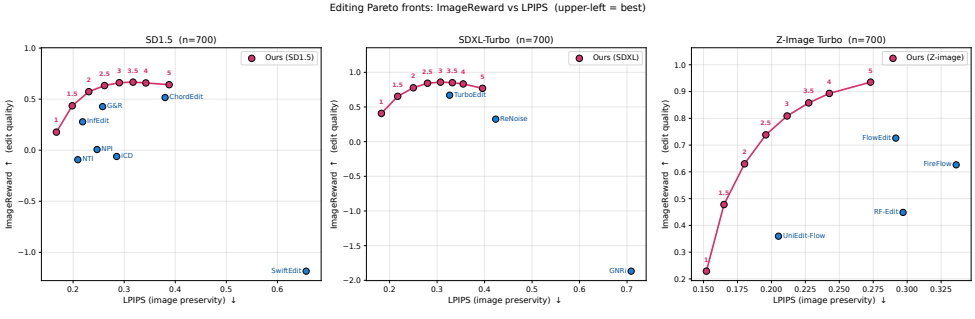


Figure 13: **Editing quality–preservation frontiers** (PIE-Bench, n=700): ImageReward↑ vs. LPIPS_{wh}(VGG-16)↓, upper-left is best. Red curve sweeps our guidance scale ω (marker labels); baselines are single points. Our frontier dominates the baselines on all three backbones (SD1.5, SDXL-Turbo, Z-Image Turbo).



Figure 14: **Additional qualitative results on Z-Image Turbo (flow-based)** against RF-Edit, FlowEdit, FireFlow, and UniEdit-Flow (one edit per row). Our method applies each edit while preserving structure, identity, and background.

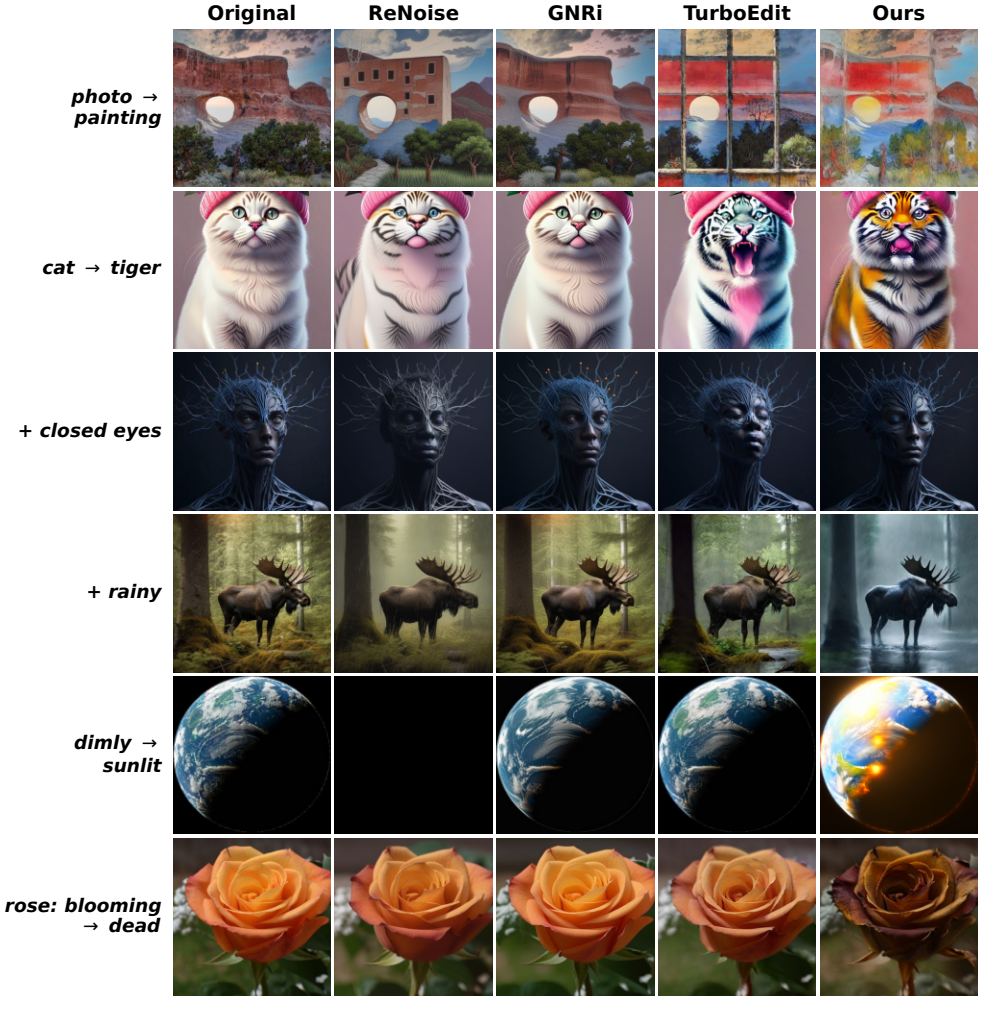


Figure 15: **Additional qualitative results on SDXL-Turbo** against ReNoise, GNRi, and TurboEdit (one edit per row).

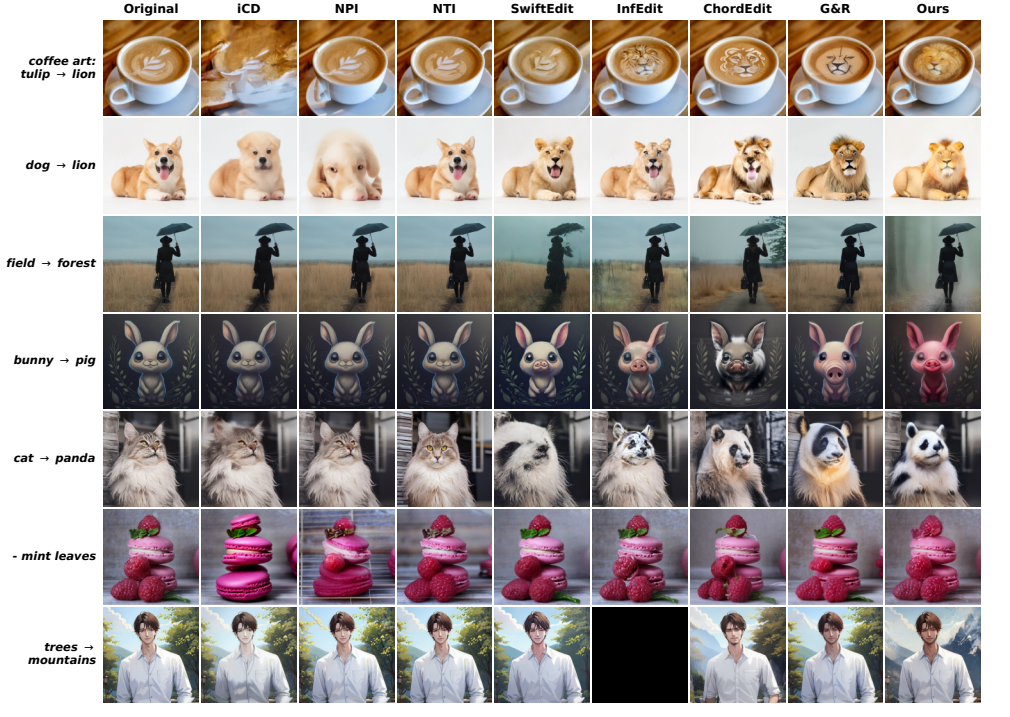


Figure 16: **Additional qualitative results on SD1.5 (LCM-DreamShaper)** against iCD, NPI, NTI, SwiftEdit, InfEdit, ChordEdit, and G&R (one edit per row).

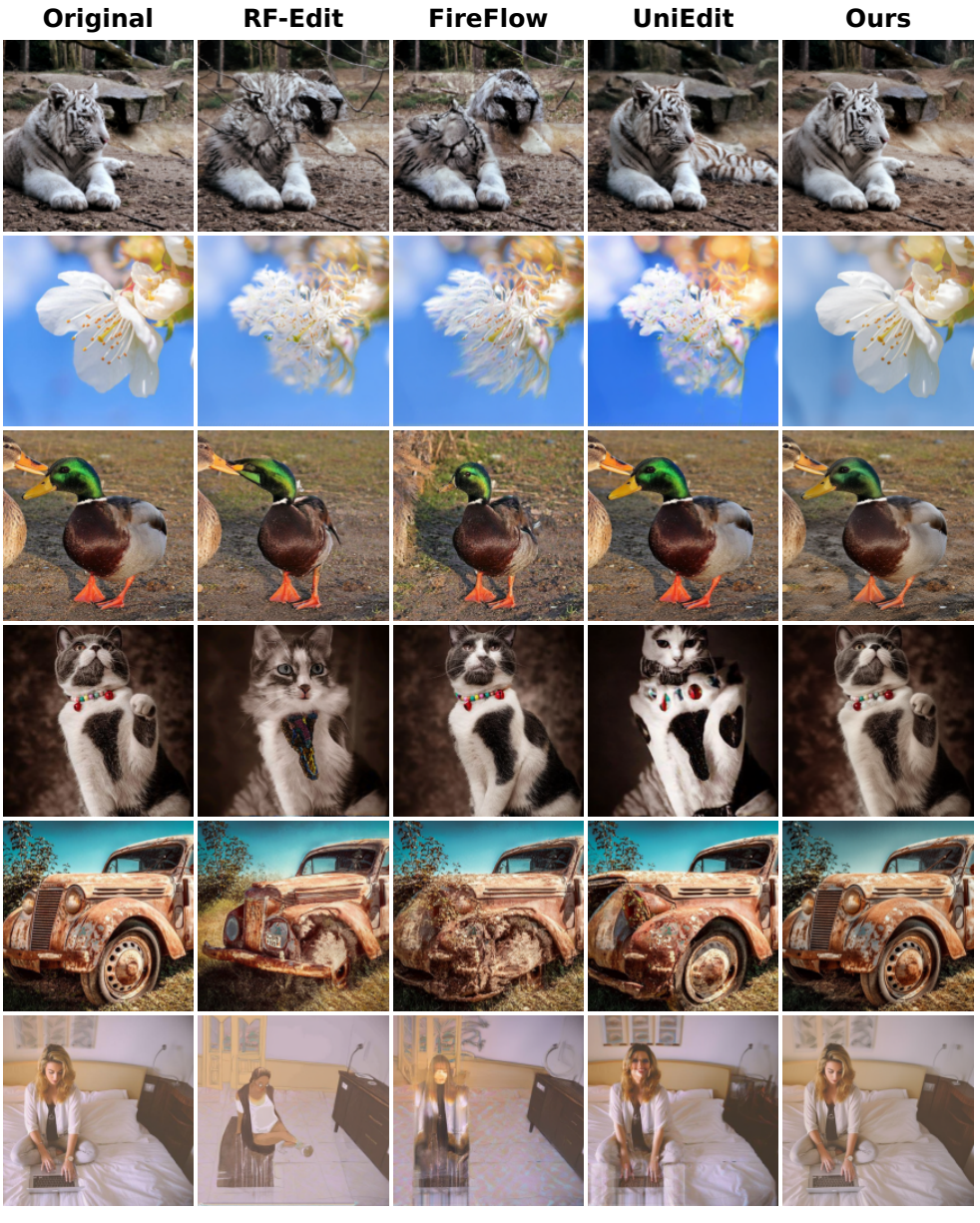


Figure 17: **Image reconstruction on Z-Image Turbo (flow-based)** against RF-Edit, FireFlow, and UniEdit-Flow. Each method inverts the source image and reconstructs it under the source prompt with classifier-free guidance disabled; a faithful method returns the input. Our adapters recover fine detail, color, and structure most closely.

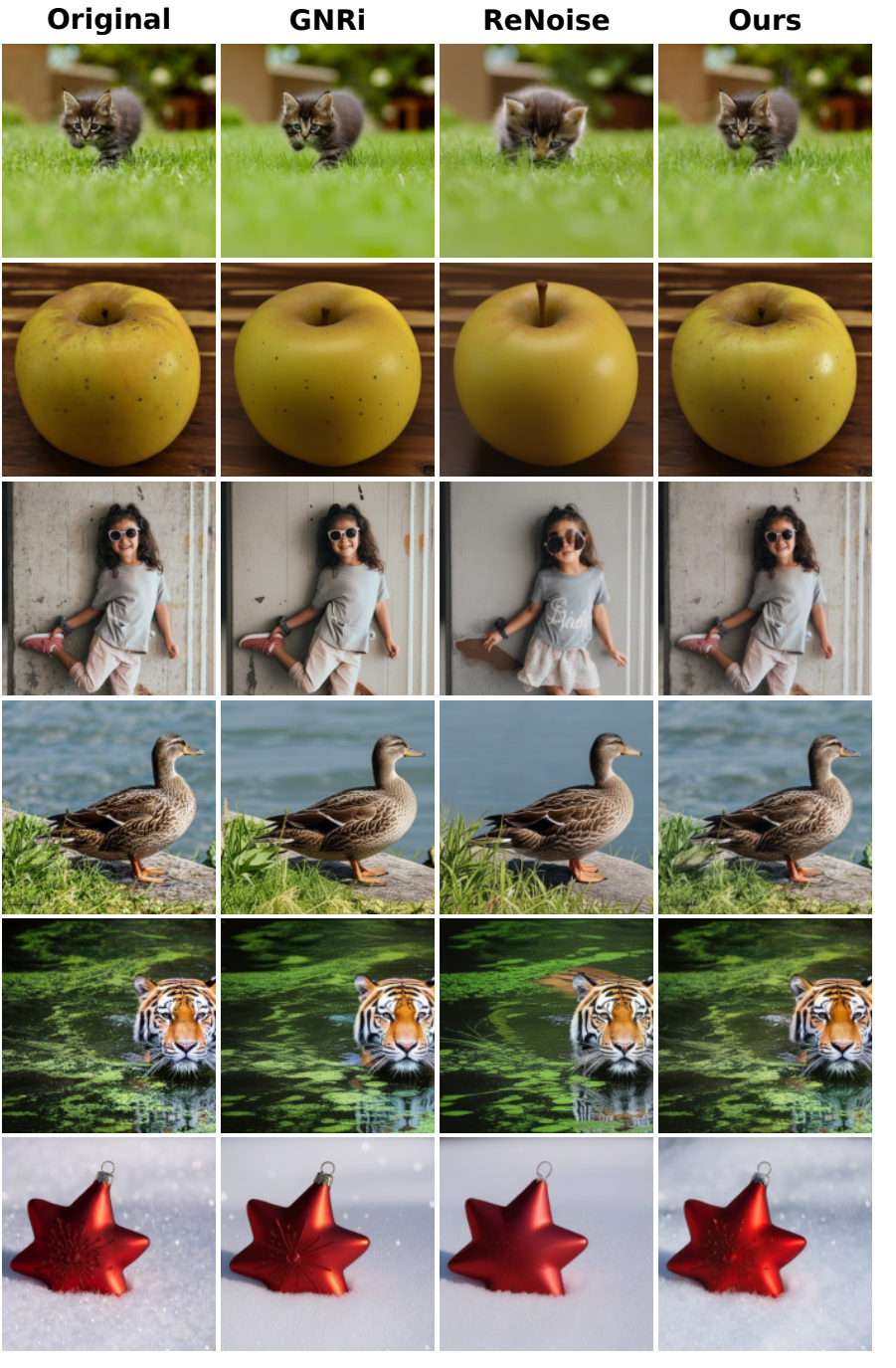


Figure 18: **Image reconstruction on SDXL-Turbo** against GNRi and ReNoise. Our method preserves identity and detail, whereas the baselines lose structure or drift in appearance.

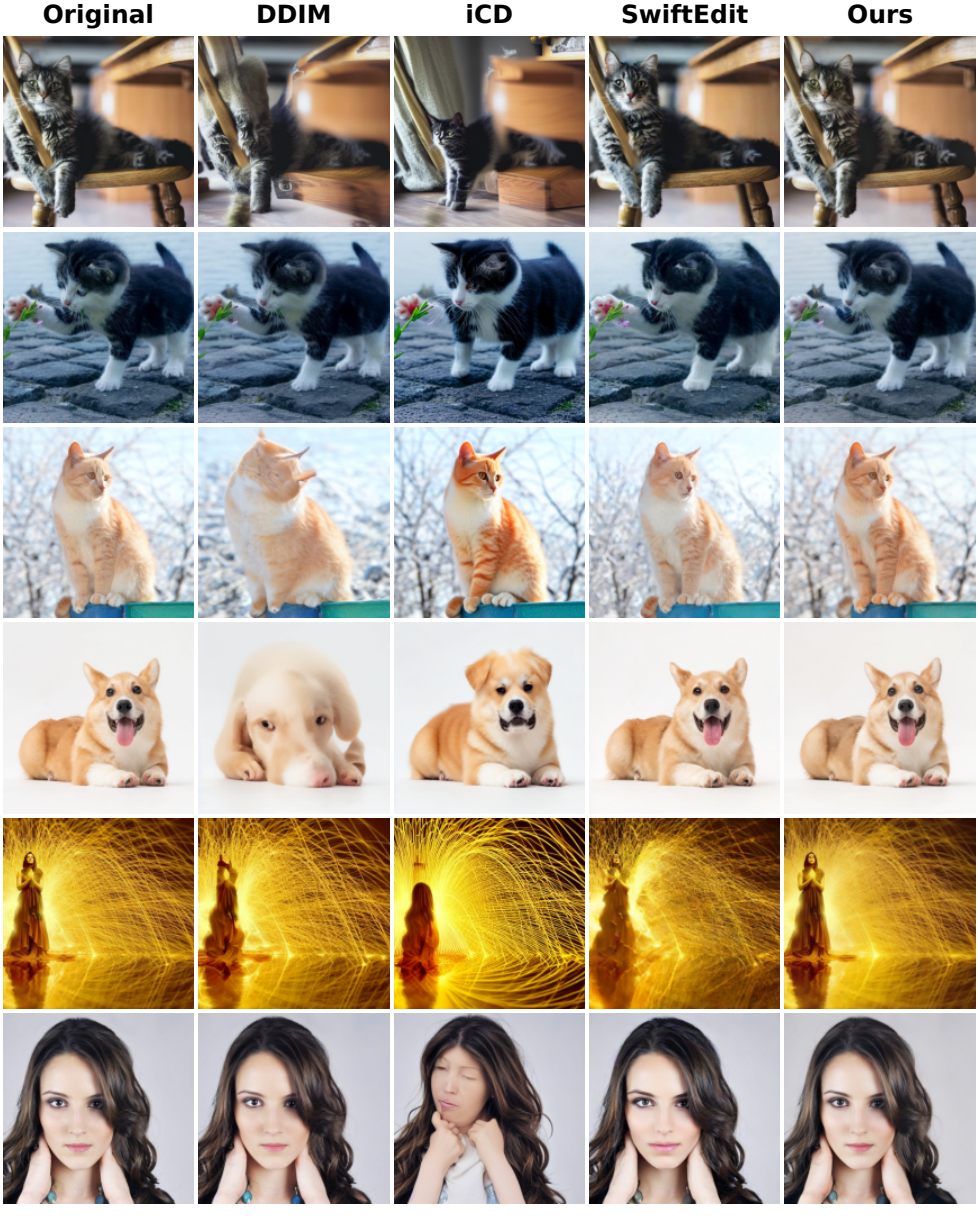


Figure 19: **Image reconstruction on SD1.5** against full-step DDIM inversion, iCD, and SwiftEdit. Our method reconstructs the input most faithfully, surpassing even full-step DDIM despite operating in the few-step regime.

Table 9: dCLIP under the training encoder (ViT-L/14) and a held-out encoder (OpenCLIP ViT-bigG/14). Within-block ordering is identical.

SD-based			*SDXL-based*			*Flow-based*		
Method	L/14	bigG	Method	L/14	bigG	Method	L/14	bigG
Ours	**0.155**	**0.183**	**Ours**	**0.178**	**0.231**	**Ours**	**0.154**	**0.193**
ChordEdit	0.119	0.145	TurboEdit	0.145	0.187	UniEdit-Flow	0.123	0.156
G&R	0.106	0.123	ReNoise	0.124	0.163	FireFlow	0.117	0.151
InfEdit	0.096	0.116	GNRi	0.038	0.043	FlowEdit	0.108	0.139
NPI	0.085	0.107				RF-Edit	0.103	0.128
SwiftEdit	0.081	0.096						
NTI	0.050	0.059						
iCD	0.047	0.055						

Table 10: Loss-weight sweeps on Z-Image-Turbo (step 600, guided inference of Eq. 23 at ω=2.5; internal scorer as in Table 1). Each row is a separately trained run; the λ=0 rows and the bold defaults coincide with the corresponding rows of Table 1.

Sweep	λ	$\text{CLIP}_{\text{wh}}\uparrow$	dCLIP↑	LPIPS↓	SSIM↑	PSNR↑	IR↑
λ_{LF} (λ_{KL}=0.1)	0	26.13	0.1423	0.2071	0.8171	21.49	0.6015
	0.25	26.66	0.1641	0.2067	0.8062	20.71	0.8302
	0.5	26.94	0.1677	0.2269	0.7922	20.21	0.8626
	1	**26.44**	**0.1541**	**0.1959**	**0.8296**	**22.01**	**0.7383**
	2	27.01	0.1714	0.2636	0.7586	18.93	0.8501
λ_{KL} (λ_{LF}=1)	0	26.06	0.1350	0.1901	0.8306	21.81	0.6029
	0.05	27.04	0.1638	0.2396	0.7804	18.73	0.8525
	0.1	**26.44**	**0.1541**	**0.1959**	**0.8296**	**22.01**	**0.7383**
	0.5	16.24	0.0561	0.5860	0.3908	8.18	−2.2740